Behaviour Under Addiction: Dynamic Choice Between Smoking and Vaping

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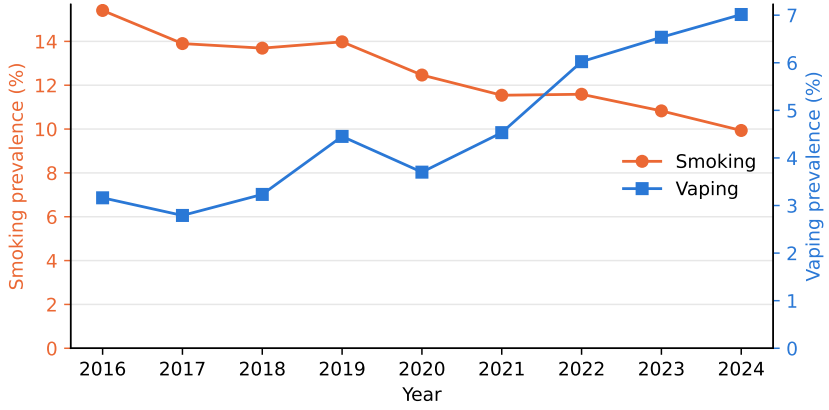
Researcher(s)' own analyses calculated (or derived) based in part on data from Nielsen Consumer LLC and marketing databases provided through the NielsenIQ Datasets at the Kilts Center for Marketing Data Center at The University of Chicago Booth School of Business.

The conclusions drawn from the NielsenIQ data are those of the researcher(s) and do not reflect the views of NielsenIQ. NielsenIQ is not responsible for, had no role in, and was not involved in analyzing and preparing the results reported herein.

Motivation

- **Health costs of cigarettes:**
  - Over $240 billion in health-care spending and about $370 billion in lost productivity in 2018 (CDC, 2024).
  - The leading cause of preventable death in the U.S.: more than 480,000 deaths a year (CDC, 2020).
  - Many policies discourage smoking. Taxes are effective and widely used (WHO, 2015).
- **E-cigarettes spread from the mid-2010s:**
  - Substitutes for combustible cigarettes, around 95% less harmful (McNeill et al., 2015), that help smokers quit (Hajek et al., 2019).
  - But addictive and not harm-free: they deliver nicotine and emit potentially toxic substances (NASEM, 2018).

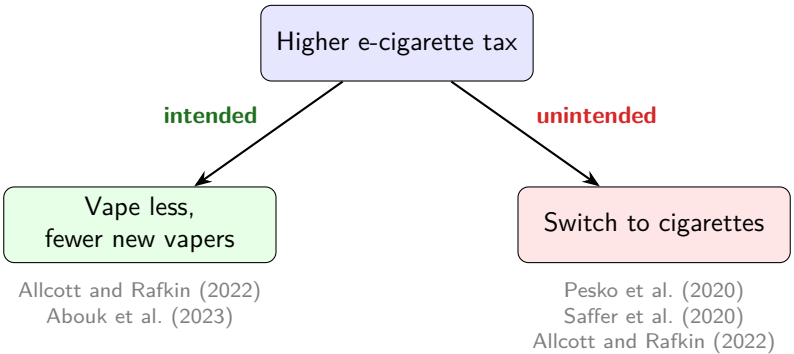
Smoking Falls, Vaping Rises



Note: Current use among U.S. adults, National Health Interview Survey (NHIS), 2016–2024. Smoking on the left axis, vaping on the right.

Motivation

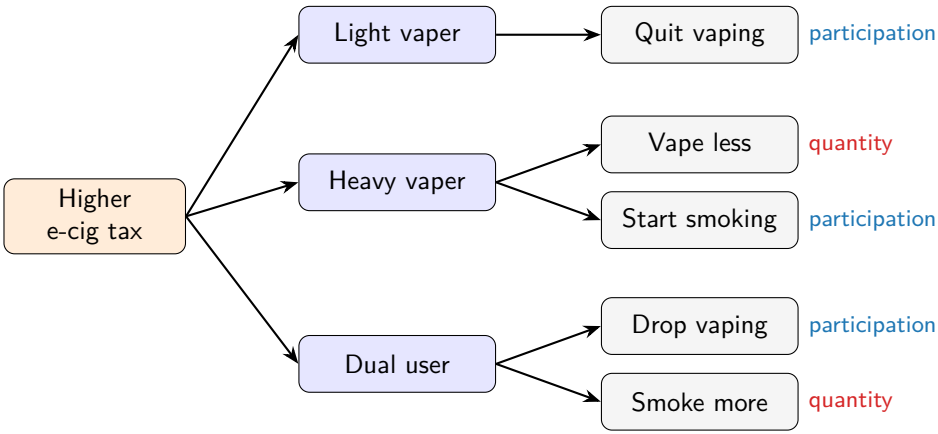
Policy makers tax "sin goods" that are enjoyable but addictive or harmful: cigarettes, alcohol, sugary drinks, and now e-cigarettes (O'Donoghue and Rabin, 2006; Gruber and Kőszegi, 2004; Allcott et al., 2019).



Motivation

- **How addiction works:** past consumption raises the reward of current consumption (Becker and Murphy, 1988).
  - Current consumption affects future consumption through addiction.
  - Requires the model to link consumption across periods.
- **Addiction changes price responses:**
  - The appeal of the good changes with addiction.
  - The cost-to-reward ratio changes with addiction.
  - Important to model addiction explicitly when measuring price responses.

Motivation: Responses Differ by Addiction and Margin



This Paper

Question: How do cigarette and e-cigarette taxes change participation and quantity choices, and how do the responses vary with addiction?

Why it matters

- Understand how tobacco demand responds to prices, on both participation and quantity.
- Measure switching between products, such as gateway effects, to guide more effective tobacco policy.
- Easily portable to study other "sin" goods.

Potential challenges

- Addiction needs to be modelled dynamically: past use shapes today's choices, and forward-looking households anticipate their own future use.
- Unobserved initial condition and price endogeneity complicate estimation.

This Paper: Approach

Model: A dynamic discrete-continuous choice model.

- Households in each period choose among {abstinence, smoking-only, vaping-only, dual-use} and how much to consume.
- Two product-specific addiction stocks accumulate with the quantity consumed.
- The addiction stocks shift the utility of participation *and* the marginal utility of quantity.

Data: NielsenIQ Consumer Panel of single-person households under 50 between 2019–2024. Prices from NielsenIQ retailer scanner data.

Methods: Maximum Likelihood.

- A control function, built from tax-instrumented price innovations, for demand shocks.
- Latent types for the unobserved pre-panel addiction stocks.
- Approximate the value function with a surrogate neural network that takes both the state variables and the parameters as inputs.

Preview of Results

1. **Addiction acts on both participation and quantity margins.**
   - Smoking: the participation utility rises at low stocks and falls at high stocks. The pull towards smoking at high stocks comes from quantity: the quantity shifter is 11 times larger at the stock ceiling.
   - Vaping: the participation utility stays positive, and desired purchases rise with the stock.
2. **Cigarettes and e-cigarettes are substitutes.**
   - Own-price elasticities: -0.6 (cigarettes), -2.6 (e-cigarettes).
   - Cross-price elasticities: 1.3 (cigarette quantity on the e-cigarette price) and 0.6 (e-cigarette quantity on the cigarette price).
3. **Addiction changes price responsiveness.**
   - Own-price responses weaken with addiction, except among the most heavily addicted users.
   - When e-cigarettes become expensive, moderately smoking-addicted users smoke more.
   - When cigarettes become expensive, smokers with little vaping addiction vape more.
4. **Bans are asymmetric (after two years).**
   - An e-cigarette ban sends a third of vapers to smoking-only, and 66% of dual-users keep smoking. Cigarette quantity $+1.5\%$.
   - A cigarette ban sends 91% of smokers to abstinence. E-cigarette quantity $+45\%$.

Related Literature

- **E-cigarette taxes & optimal taxation of sin goods:**
  (Pesko et al., 2020; Saffer et al., 2020; Pesko and Warman, 2022; O'Donoghue and Rabin, 2006; Farhi and Gabaix, 2020; Allcott and Rafkin, 2022)
  – I separate participation and quantity choices, and show the responses depend on addiction.

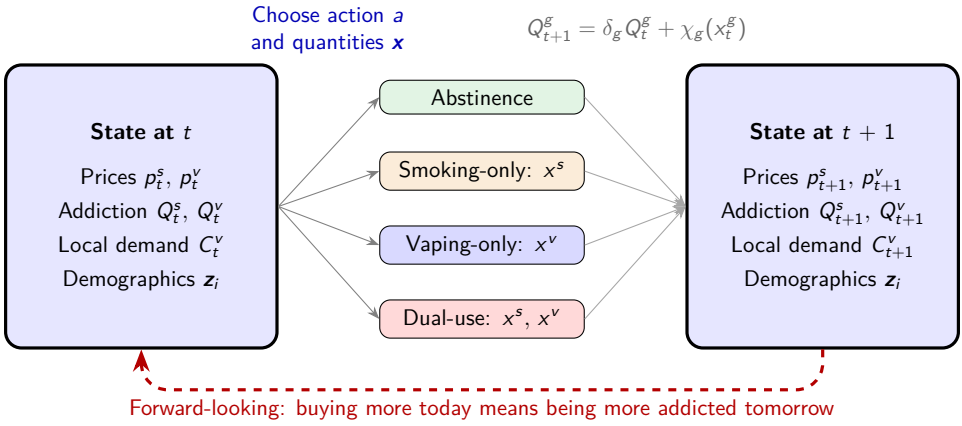
- **Rational addiction & multiple addictions:**
  (Becker and Murphy, 1988; Chaloupka, 1991; Gruber and Kőszegi, 2001; Bask and Melkersson, 2003; Pierani and Tiezzi, 2009; Gordon and Sun, 2015; Chen and Rao, 2020)
  – I add product-specific addictions to allow for richer switch pattern.

Related Literature

- **Participation & quantity (discrete–continuous demand):**
  (Cragg, 1971; Jones, 1989; Dubin and McFadden, 1984; Hanemann, 1984; Erdem et al., 2003; Hendel and Nevo, 2006)
  – I adopt a dynamic discrete-continuous choice model, with two product-specific addiction stocks, and latent types

- **Neural-network estimation:**
  (Norets, 2012; Maliar et al., 2021; Azinovic et al., 2022; Kase et al., 2022; Friedl et al., 2023; Chen et al., 2026; Scheidegger, 2026)
  – I integrate a surrogate neural network into solving households' dynamic problem.

The Model in One Picture



Flow Utility

The utility for action a, which consumes product $g \in \mathcal{G}(a)$ is:

$$u_{it,a}(\boldsymbol{x}) = \underbrace{\boldsymbol{z}_i'\gamma_a + \zeta_{t,a} + \mu_{c,a} + f_a(Q^s_{it}, Q^v_{it})}_{\text{participation}} + \underbrace{\sum_{g \in \mathcal{G}(a)} \left[\Phi^g_{it}\, \phi_g(x^g; \kappa_{git}) - \alpha\, p^g_{ct}\, x^g \right]}_{\text{quantity}} + \varepsilon_{iat}$$

- E.g. when smoking ($a = 1$), households consume only cigarettes $g \in \{\text{cigarettes}\}$.
- Utility function coefficients are action-specific.
- To simplify notation, I use smoking as an example in the following (dropping a and g).

Flow Utility

$$u_{it}(x) = \underbrace{\boldsymbol{z}_i'\gamma + \zeta_t + \mu_c + f(Q^s_{it}, Q^v_{it})}_{\text{participation}} + \underbrace{\Phi_{it}(Q^s_{it}) \cdot \phi(x; \kappa_{it}) - \alpha\, p_{ct}\, x}_{\text{quantity}} + \varepsilon_{it}$$

- Demographics $\boldsymbol{z}_i$, month effects ζ_t, county effects μ_c, ε_{it} follows i.i.d. EV1.
- Abstinence is the reference action: its utility is normalised to zero.

Flow Utility

$$u_{it}(x) = \underbrace{\boldsymbol{z}_i'\gamma + \zeta_t + \mu_c + f(Q^s_{it}, Q^v_{it})}_{\text{participation}} + \underbrace{\Phi_{it}(Q^s_{it}) \cdot \phi(x; \kappa_{it}) - \alpha\, p_{ct}\, x}_{\text{quantity}} + \varepsilon_{it}$$

- Quantity utility:

$$\Phi_{it}(Q^s_{it}) \cdot \phi(x; \kappa) = \exp(\psi_0 + \psi\, Q^s_{it}) \times \frac{(x+\gamma)^{1-\kappa(Q^s_{it})} - \gamma^{1-\kappa(Q^s_{it})}}{1-\kappa(Q^s_{it})}$$

  where $\kappa(Q^s_{it}) = \kappa_0 + \kappa_1 Q^s_{it}$. Static elasticity $\approx -1/\kappa(Q^s_{it})$.
- Recorded purchase $x_{it} = x^*_{it}\, e^{\nu_{it}}$. $\nu_{it} \sim N(0, \sigma^2)$ is the observation error (purchase vs. consumption, aggregation, recording).

Addiction Dynamics

Product-specific addiction stock:

$$Q^s_{i,t+1} = \delta_s \, Q^s_{it} + \chi(x_{it}), \qquad \text{(e-cigarettes: same form, with } \delta_v\text{)}$$

- $\chi(x)$ is increasing in quantity x.

Why two separate stocks:

- Nicotine is common in both.
- Flavour, throat hit and ritual are product-specific.
- A chronic smoker may still find vaping appealing through nicotine, but less so than a vaper with equivalent nicotine addiction.

χ

Dynamic Problem and Quantity Choice

Value of smoking is:

$$V_{a=1}(\boldsymbol{S}_{it}, \boldsymbol{z}_i) = \max_{x \geq 0} \left\{ \bar{u}_{it}(x) + \beta \, \mathbb{E}\big[V(\boldsymbol{S}_{i,t+1}, \boldsymbol{z}_i) \mid \boldsymbol{S}_{it}, s, x \big] \right\}$$

The F.O.C. requires:

$$\underbrace{\Phi_{it} \, \phi'\big(x^*; \kappa_{it}\big)}_{\text{marginal utility}} = \underbrace{\alpha p_{ct}}_{\text{money cost}} \underbrace{- \, \beta \, \Lambda \, \chi'(x^*)}_{\text{value of a larger stock tomorrow}}$$

where $\Lambda = \partial \, \mathbb{E}[V(\boldsymbol{S}_{t+1}) \mid \boldsymbol{S}_{it}, s, x^*] / \partial Q^s_{t+1}$.

- Consuming more today raises tomorrow's stock. A forward-looking household prices this in through Λ.

Demand Shocks

Factors like social norms shifting, viral trends and advertising creates persistent demand shocks ξ_{ct}:

$$u_{it} = \bar{u}_{it} + \xi_{ct} + \varepsilon_{it} \qquad \text{– change the participation utility}$$
$$\log \Phi_{it} = \psi_0 + \psi\, Q^s_{it} + b\, \xi_{ct} \qquad \text{– change the quantity utility}$$

- **Price endogeneity:** $\mathrm{Cov}(\boldsymbol{p}_{ct}, \boldsymbol{\xi}_{ct}) \neq 0$, the price coefficient is biased.
- **Residual distribution:** $\xi + \varepsilon$ is no longer EV1.
- **Addiction Stocks:** $\mathrm{Cov}(Q_{it}, \boldsymbol{\xi}_{ct}) \neq 0$, the stock coefficient is biased.
- **Expectations:** V depends on ξ. A positive shock today amplifies through tomorrow's value.

The Control Function

Construction:

1. Regress price on state taxes $p_{ct} = B\,\boldsymbol{\tau}_{st} + \cdots + v_{ct}$. Predict the unexplained residual $\hat{\boldsymbol{v}}_{ct}$.
2. The residual $\hat{\boldsymbol{v}}_{ct}$ carries the demand shocks.
3. Iterating backwards, since demand shocks are persistent: $\widehat{C}_{ct} = \rho\,\widehat{C}_{c,t-1} + \boldsymbol{\lambda}'\hat{\boldsymbol{v}}_{ct}$.

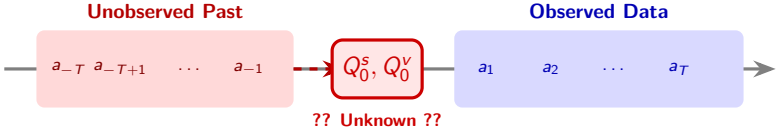
The fitted $\widehat{C}_{ct}$ replaces the shock on two margins:

1. **Participation utility**: $u_{it} = \bar{u}_{it} + \widehat{C}_{ct}$.
2. **Quantity shifter**: $\log \Phi_{it} = \psi_0 + \psi\, Q^s_{it} + b\,\widehat{C}_{ct}$.

assumptions first stage estimates

Unobserved Initial Stocks

Initial addiction Q_0^s, Q_0^v are never directly observed — they accumulate from past consumption.

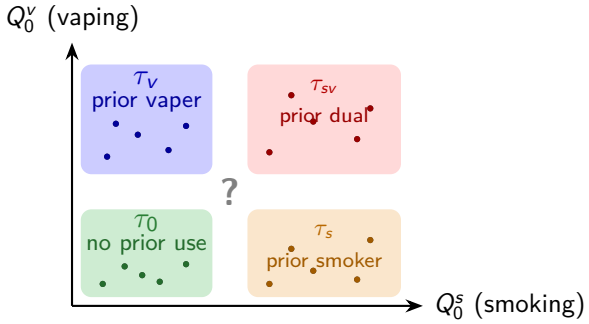


Why it matters:

- The estimation depends on the *current* addiction stock.
- Wrong $Q_0 \Rightarrow$ wrong state path $\Rightarrow$ biased estimates.

Four Latent Addiction Types

Four types indexed by initial addiction stocks (Q_0^s, Q_0^v):



- Types differ in initial addiction capital (Q_0^s, Q_0^v), and the action-specific intercepts.
- Type probabilities are a multinomial logit on demographics.

type definitions

The Likelihood

Each household's per-latent-type likelihood is:

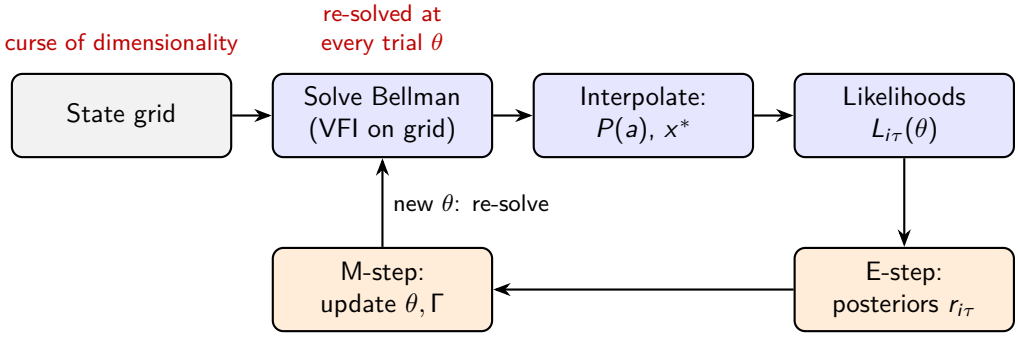
$$L_{i\tau}(\theta) = \prod_{t \in \mathcal{T}_i} \underbrace{P_\tau\big(a_{it} \mid \boldsymbol{S}^\tau_{it}(\theta), \boldsymbol{z}_i\big)}_{\text{which products}} \underbrace{\frac{1}{\sigma\, x_{it}}\, n\Big(\frac{\log x_{it} - \log x^*_{i\tau t}(\theta)}{\sigma}\Big)}_{\text{how much}}$$

with $P_\tau\big(a_{it} \mid \boldsymbol{S}^\tau_{it}(\theta), \boldsymbol{z}_i\big) = \frac{\exp(V_a)}{\sum_{a'} \exp(V_{a'})}$, and $\nu_{it} = \log x_{it} - \log x^*_{i\tau t}(\theta) \sim N(0, \sigma^2)$.

The total likelihood is:

$$\mathcal{L}(\theta, \Gamma) = \sum_i w_i \log \Big[\sum_{\tau \in \mathcal{T}} \underbrace{\pi_\tau(\boldsymbol{z}_i; \Gamma)}_{\text{type probability}} \; L_{i\tau}(\theta) \Big]$$

The Conventional Estimation Route



Problem:

- **Curse of dimensionality:** state space approaching $\approx 10^{11}$.
- **Computationally intense:** re-solve Bellman equation at every θ value.

Deep Equilibrium Network

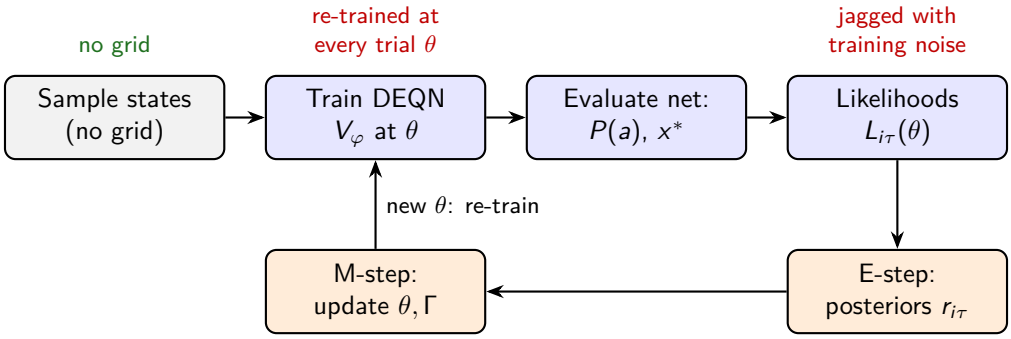
For a given θ, train a network $V_\varphi(\boldsymbol{S})$ to match the Bellman target (Azinovic et al., 2022):

$$\mathcal{T}_\varphi(\boldsymbol{S};\theta) = \log \sum_{a\in\mathcal{A}} \exp\left\{ \max_{\boldsymbol{x}_a \geq 0} \left[\bar{u}_a(\boldsymbol{S}, \boldsymbol{x}_a) + \beta\, \mathbb{E}[V_\varphi(\boldsymbol{S}';\theta) \mid \boldsymbol{S}, a, \boldsymbol{x}_a]\right]\right\}$$

Advantages:

1. No grid needed.
2. Value function is accurate at the variable values instead of only at grid points.

The Estimation Loop with a Deep Equilibrium Network



New problems:

- **Computationally intense:** costly to re-train the net at every θ value.
- **Re-training noise:** each time the re-trained net introduces errors into the likelihood, making the gradients inaccurate (Moré and Wild, 2012).

Surrogate Neural Network

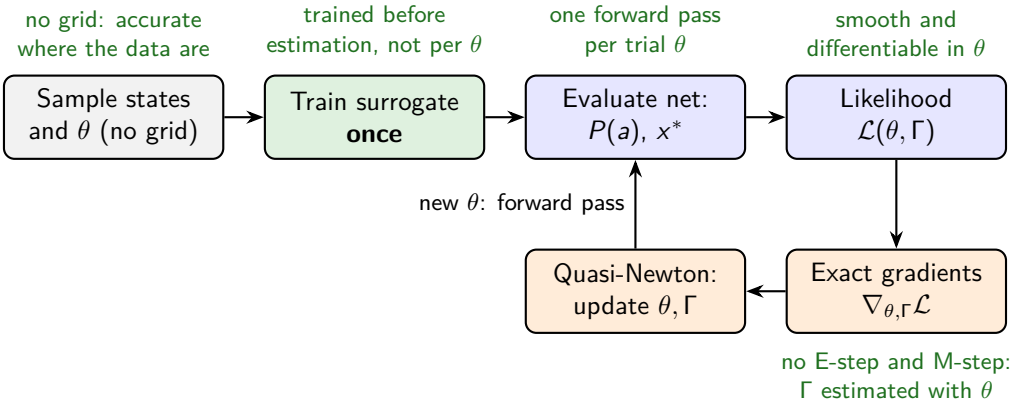
Treat θ as additional state variables and train **one** network with states and parameters together (Norets, 2012; Kase et al., 2022; Chen et al., 2026):

$$V(\boldsymbol{S}, \theta) = \log \sum_{a \in \mathcal{A}} \exp \left\{ \max_{\boldsymbol{x}_a \geq 0} \left(\bar{u}_a(\boldsymbol{S}, \boldsymbol{x}_a; \theta) + \beta \, \mathbb{E}\left[V(\boldsymbol{S}', \theta) \mid \boldsymbol{S} \right] \right) \right\} + \gamma_E$$

In addition, a policy net $x_{\mathrm{NN}}(\boldsymbol{S}, \theta)$ is also trained to meet the first-order condition for quantity.

After training, the network weights are fixed. Solving the model at a new θ is **one forward pass**.

Updated Estimation Loop

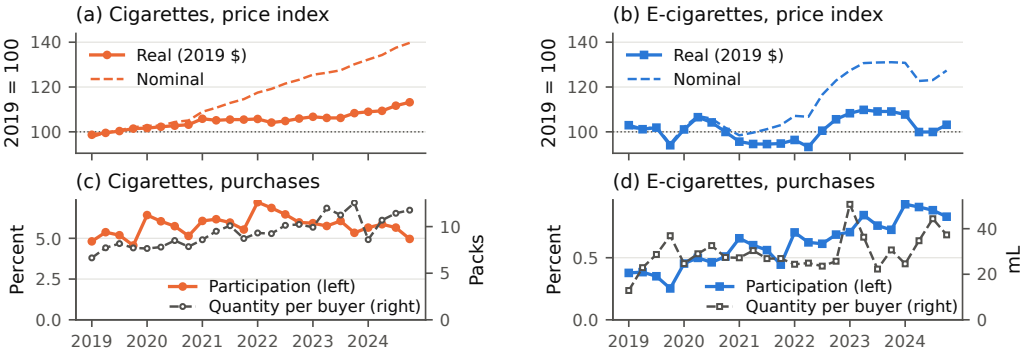


Data and Sample

- Purchases: NielsenIQ Consumer Panel (NielsenIQ, 2024a), UPC-level purchases and demographics of about 60,000 households a year.
- Prices: NielsenIQ Retail Scanner (NielsenIQ, 2024b), weekly store-UPC sales, aggregated to county-month prices per pack and per mL. price index
- Taxes: state cigarette taxes from Orzechowski and Walker (2024). E-cigarette taxes from Cotti et al. (2023, 2026).

sample construction identification

Smoking and Vaping in the Panel



- Cigarette real price rising steadily, e-cigarette price volatile during sample period.
- Vaping participation more than doubles over the sample (0.34% to 0.89%).

more descriptives underreport

Behaviour is Persistent

	Action in t (%)			
Action in $t-1$	Abstinence	Smoking-only	Vaping-only	Dual-use
Abstinence	**98.3**	1.5	0.2	0.0
Smoking-only	26.1	**72.6**	0.3	0.9
Vaping-only	34.1	1.2	**61.8**	2.9
Dual-use	20.7	41.1	12.3	**26.0**

- Smokers smoke again next month 73% of the time, vapers vape again 62%.
- Direct switches between the products are rare: 0.3% of smokers and 1.2% of vapers.
- Dual-use is the least stable: only 26% buy both again, 41% keep cigarettes only.

Estimates: Participation

Addictions (Q^s, Q^v) enter the utility function as:

$$u_{it,a}(x) = f_{s1,a}Q^s + f_{s2,a}(Q^s)^2 + f_{v1,a}Q^v + f_{v2,a}(Q^v)^2 + ...$$

Participation:

	Smoking-only	Vaping-only	Dual-use
$\hat{f}_{s1,a}$	1.090 (0.063)	−0.878 (0.371)	0.571 (0.487)
$\hat{f}_{s2,a}$	−1.961 (0.101)	0.022 (0.490)	−1.750 (0.361)
$\hat{f}_{v1,a}$	−0.225 (0.059)	0.949 (0.171)	−0.034 (0.139)
$\hat{f}_{v2,a}$	0.136 (0.142)	−0.744 (0.145)	0.034 (0.130)

- Own stocks raise the utility of the own action. The negative squared terms are a self-perceived health cost that grows with addiction (inverse U).
- The cigarette stock *lowers* the utility of vaping-only; the vaping stock lowers the utility of smoking-only slightly.

participation utility by stock

Estimates: Quantity

Quantity x_{it} enters the utility function as:

$$u_{it}(x) = (\text{participation terms}) + \Phi_{it}\,\phi(x_{it};\kappa_{it}) - \alpha\, p_{ct}\, x_{it} + \varepsilon_{it}$$

$$\Phi_{it} = \exp(\psi_0 + \psi\, Q^s_{it}), \quad \phi(x;\kappa) = \frac{(x+\gamma)^{1-\kappa} - \gamma^{1-\kappa}}{1-\kappa}, \quad \kappa_{it} = \kappa_0 + \kappa_1 Q^s_{it}$$

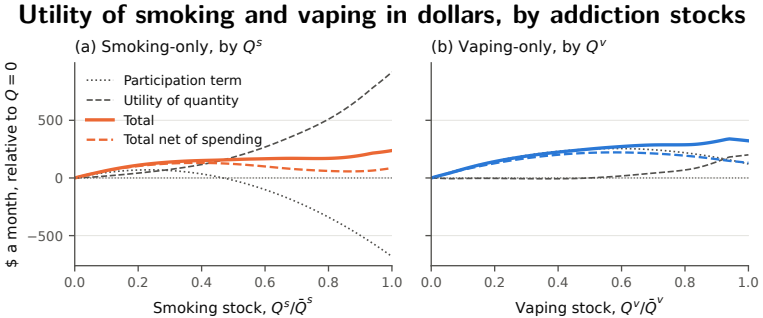
Quantity: utility of quantity

	Cigarettes	E-cigarettes
Shifter constant ψ_0	−1.212 (0.112)	0.083 (0.236)
Shifter, own stock ψ	0.280 (0.011)	−0.042 (0.016)
Curvature constant κ_0	1.180 (0.059)	1.129 (0.125)
Curvature, own stock κ_1	−0.163 (0.058)	−0.647 (0.124)
Money coefficient α	1.971 (0.157)	

- The cigarette shifter rises with the stock: 11 times larger at the stock ceiling. The e-cigarette shifter falls with the stock.
- Curvature falls with the stock, most for e-cigarettes (κ_v from 1.13 to 0.48).

marginal value by stock | control function estimates | latent types

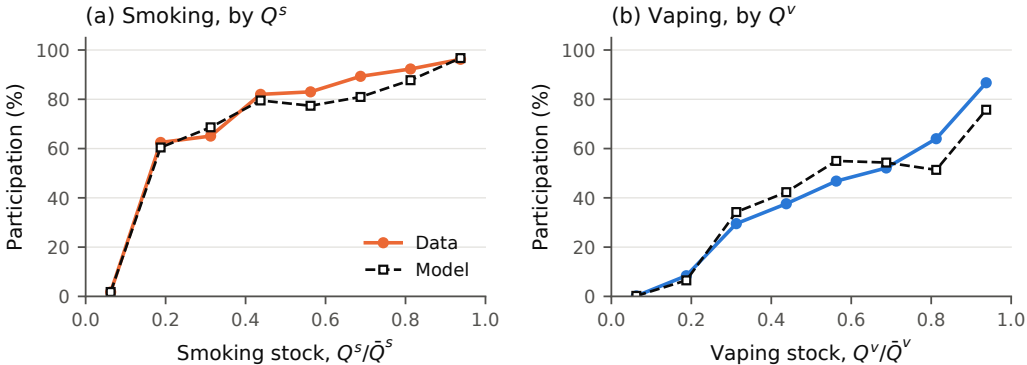
Stocks and Utility: Total



- **Smoking:** at the stock ceiling the participation term is $680 a month below its zero-stock level, the utility of quantity $920 above it: the total is $240 higher.
- The household perceives the health cost, but the large quantity keeps it smoking.
- **Vaping:** the participation term stays positive. The total rises to $320 a month at the ceiling.

Note: Utility of quantity at the model's predicted purchase in each stock bin; spending at the mean price.

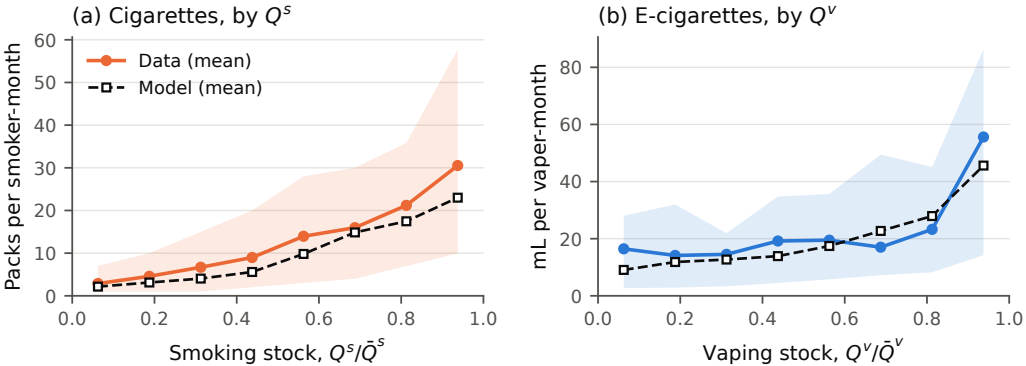
Model Fit: Participation



- Overall shares match: 93.8% abstinence and 5.6% smoking-only, against 93.7% and 5.7% in the data.
- Smoking fits from almost no stock to mild addiction and is slightly under-predicted at medium to high stocks. Vaping rises with the stock as in the data.

calibration, by period, dual-use, dynamic fit

Model Fit: Quantity



- Predicted purchases rise with the stock, as in the data, but stay below it: 6.9 against 9.3 packs and 23.5 against 29.2 mL per buyer-month.
- E-cigarettes: slightly too low at low stocks and slightly too high at higher stocks.

Policy Experiments

Three policy changes, one product at a time:

1. A permanent **5% tax** (about $0.36 per pack, $0.26 per mL). Households expect it to persist.
2. A **ban**: the actions that include the product are removed from the choice set.
3. **Removing** the average state excise tax: $1.84 per pack, $0.31 per mL.

Immediate response at the current stocks (month 0), then 24 months of simulation as the stocks adjust, in a simulated population that covers every combination of the two stocks.

simulated population

Headline Elasticities

Permanent 5% tax, elasticity per 1%, stocks at current values, total quantity per capita.

Price raised by 1%	Cigarette quantity **total**	E-cigarette quantity **total**
Cigarette price	$\mathbf{-0.61}$	$\mathbf{+0.60}$
E-cigarette price	$\mathbf{+1.26}$	$\mathbf{-2.63}$

- **Substitutes**, with cross-price elasticities of 1.3 and 0.6. They are less precise than the own-price ones.
- The own-price elasticity of cigarettes is slightly larger in magnitude than the literature's -0.3 to -0.5 for total adult demand (Chaloupka and Warner, 2000; Pesko et al., 2020).

Headline Elasticities: Own-Price Margins

Permanent 5% tax, elasticity per 1%, stocks at current values, total quantity per capita.

Price raised by 1%	Cigarette quantity **total** (partic., quantity)	E-cigarette quantity **total** (partic., quantity)
Cigarette price	**−0.61** (−0.10, −0.51)	**+0.60**
E-cigarette price	**+1.26**	**−2.63** (−1.06, −1.65)

- Cigarettes respond mostly through the quantity margin (−0.51 of −0.61).
- E-cigarettes respond on both margins, quantity per user slightly more (−1.65 of −2.63).

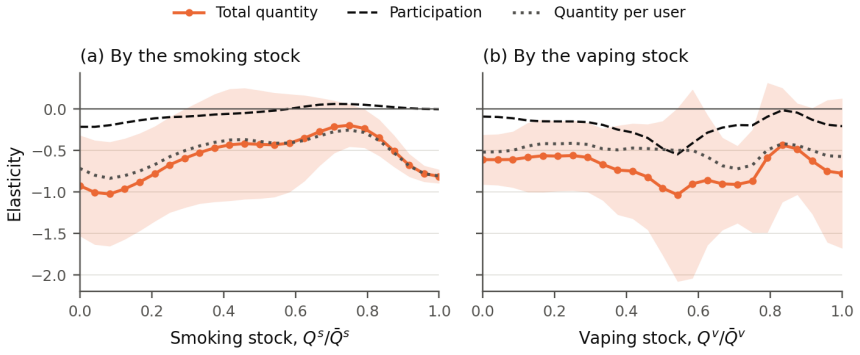
Headline Elasticities: Cross-Price Margins

Permanent 5% tax, elasticity per 1%, stocks at current values, total quantity per capita.

Price raised by 1%	Cigarette quantity **total** (partic., quantity)	E-cigarette quantity **total** (partic., quantity)
Cigarette price	**−0.61** (−0.10, −0.51)	**+0.60** (+0.39, +0.21)
E-cigarette price	**+1.26** (+0.32, +0.92)	**−2.63** (−1.06, −1.65)

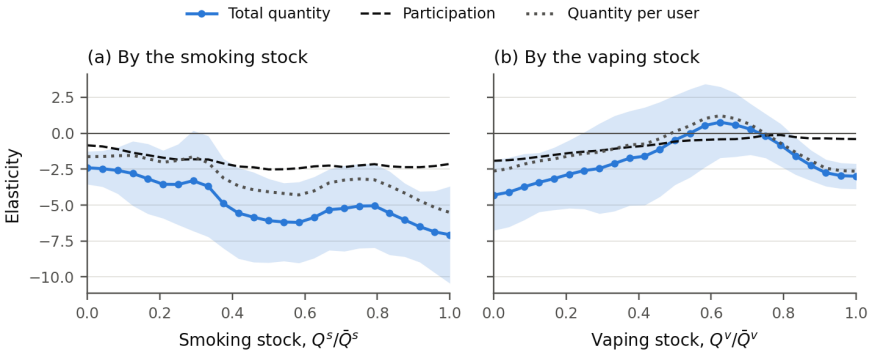
- Higher e-cigarette prices lead smokers to **smoke more**, mostly through quantity (+0.92 of +1.26).
- Higher cigarette prices raise e-cigarette use more through participation (+0.39 of +0.60).

Own-Price Elasticity of Cigarettes, by Stock



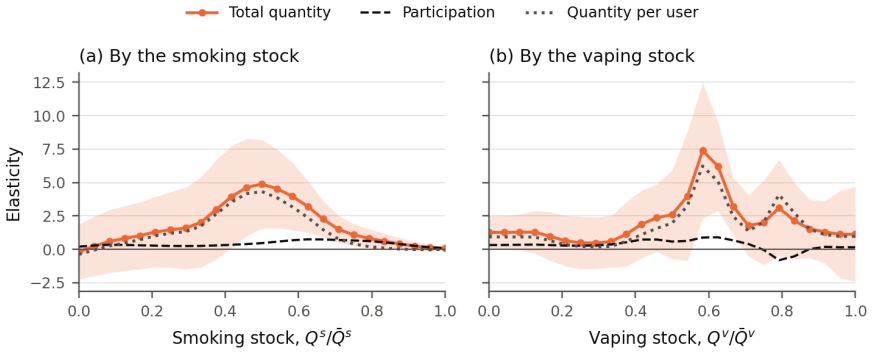
- By the smoking stock: -0.9 with no cigarette stock, smallest around three quarters of the ceiling, and -0.8 again near the ceiling, where the large purchases of the most heavily smoking-addicted users respond to the price.
- By the vaping stock: stable at around -0.7.

Own-Price Elasticity of E-Cigarettes, by Stock



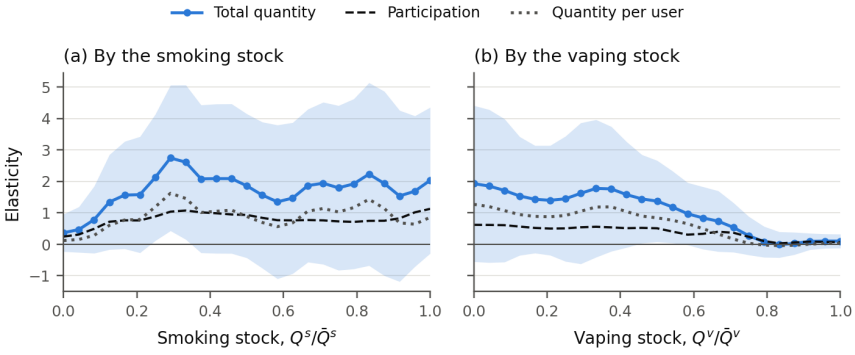
- By the vaping stock: -4.3 with no vaping stock, about zero at middle addiction levels, and -3.0 near the ceiling, mostly through quantity per user.
- By the smoking stock: larger in magnitude as the stock grows, to -7.1 near the ceiling. Smoking-addicted users vape mostly as dual-users, and their e-cigarette purchases are very price-sensitive.

Cigarette Quantity and the E-Cigarette Price, by Stock



- By the smoking stock: close to zero with no cigarette stock and near the ceiling, largest around half the ceiling, mostly through quantity per user. Moderately smoking-addicted users smoke more when e-cigarettes become more expensive.
- By the vaping stock: around 1.2, peaking a little above half the ceiling.

E-Cigarette Quantity and the Cigarette Price, by Stock



- By the smoking stock: 0.4 with no cigarette stock, around 2.0 from lightly to heavily smoking-addicted users. Smokers are the ones who vape more.
- By the vaping stock: 1.9 with no vaping stock, fading to 0.1 near the ceiling. Heavily vaping-addicted users do not respond.

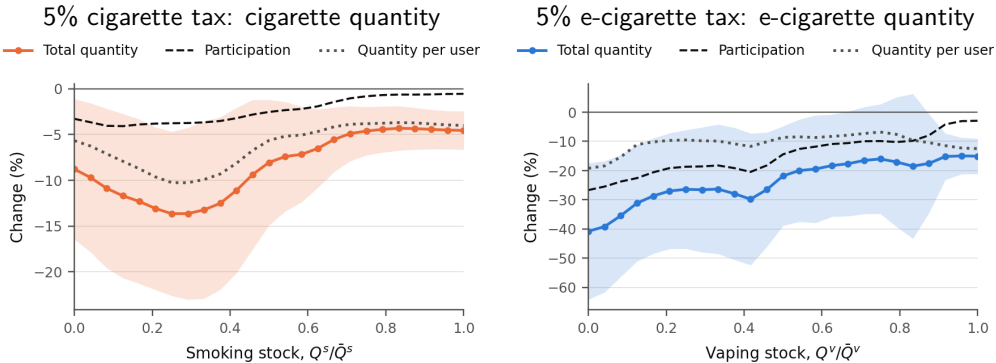
Letting the Stocks Adjust: 24 Months

Permanent 5% tax, percent change in total quantity per capita.

5% tax on	Quantity of	First month	After 24 months
Cigarettes	cigarettes	-3.0%	-7.1%
E-cigarettes	e-cigarettes	-13.1%	-29%
E-cigarettes	cigarettes	$+6.3\%$	$+29\%$
Cigarettes	e-cigarettes	$+3.0\%$	$+13\%$

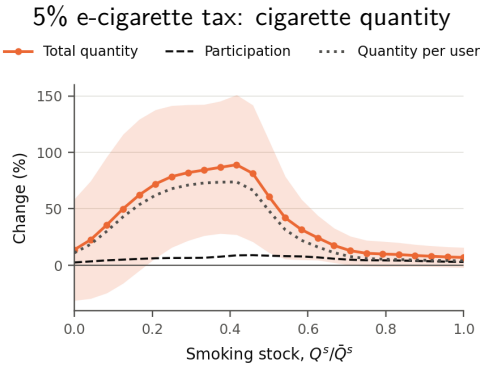
- Responses grow as the stocks adjust: smaller purchases lower the stock, which lowers later purchases and participation.
- The cross effect on cigarettes grows most: households that smoke more build a larger cigarette stock, which raises the marginal utility of every further pack.

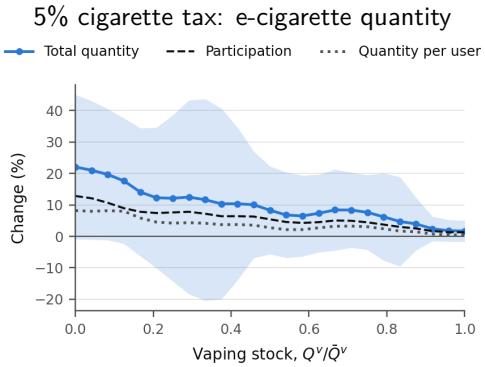
After 24 Months: Own-Price Responses, by Stock



- Cigarette tax: cigarette quantity falls by 8.8% with no cigarette stock and by 4.6% near the ceiling, most among lightly smoking-addicted users and mostly through quantity per user.
- E-cigarette tax: e-cigarette quantity falls by 41% with no vaping stock and by 15% near the ceiling, mostly through participation.

After 24 Months: Cross-Price Responses, by Stock





- E-cigarette tax: cigarette quantity rises by 14% with no cigarette stock and by 7% near the ceiling. Moderately smoking-addicted users, whose purchases and stocks have room to grow, smoke more.
- Cigarette tax: e-cigarette quantity rises by 22% with no vaping stock and by 1.7% near the ceiling, mostly through participation.

by the other product's stock

E-Cigarette Ban

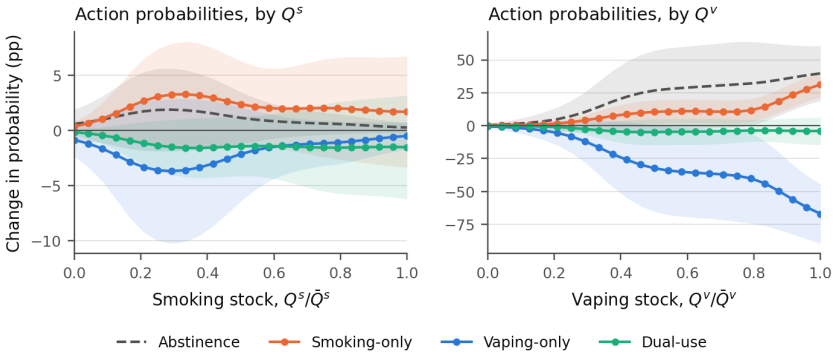
Action shares after 24 months (% of households)

	Baseline	E-cig. ban	Cig. ban	No cig. tax	No e-cig. tax
Abstinence	94.65	95.17	98.79	93.73	94.38
Smoking-only	4.50	4.83	–	5.72	4.64
Vaping-only	0.72	–	1.21	0.46	0.84
Dual-use	0.13	–	–	0.09	0.15

- Of the households that would vape alone at month 24, 67% move to abstinence and 33% to smoking-only. 66% of would-be dual-users keep smoking.
- Smoking participation +4.3%, but cigarette quantity only +1.5%: the former vapers who smoke buy fewer packs than existing smokers.

first month purchase sizes by action

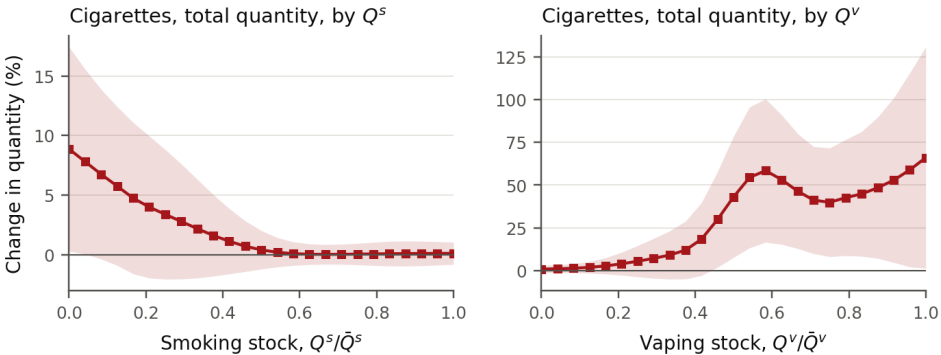
E-Cigarette Ban: Who Switches, by Stock



- The response grows with the vaping stock: near the ceiling, about 40 p.p. go to abstinence and 30 p.p. to smoking-only.
- By the cigarette stock the changes are small.

Note: Changes 24 months after the ban, by the stocks at the time of the ban, against the baseline at month 24.

E-Cigarette Ban: Cigarette Quantity, by Stock



- Cigarette quantity rises with the vaping stock, by about 65% near the ceiling.
- By the cigarette stock the changes are small.

Cigarette Ban

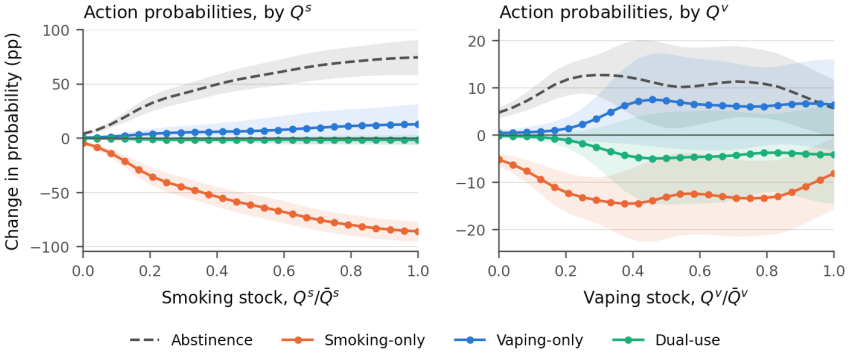
Action shares after 24 months (% of households)

	Baseline	E-cig. ban	Cig. ban	No cig. tax	No e-cig. tax
Abstinence	94.65	95.17	98.79	93.73	94.38
Smoking-only	4.50	4.83	–	5.72	4.64
Vaping-only	0.72	–	1.21	0.46	0.84
Dual-use	0.13	–	–	0.09	0.15

- Of the households that would smoke alone at month 24, 91% move to abstinence and 9% to vaping-only. 37% of would-be dual-users keep vaping.
- Vaping participation +42% and e-cigarette quantity +45%, from a small base.

first month purchase sizes by action

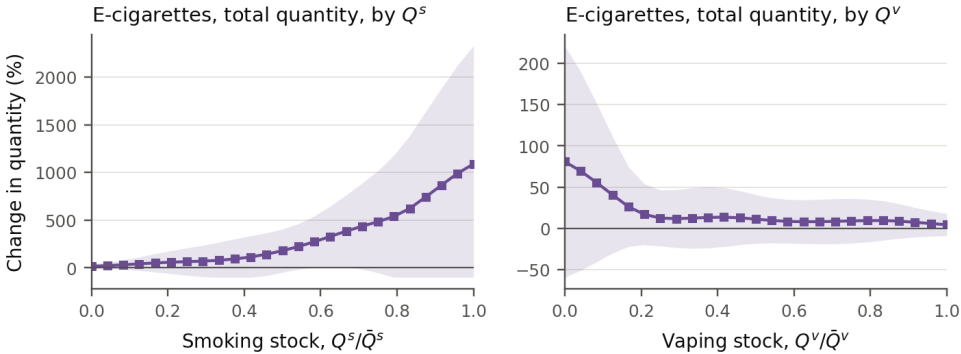
Cigarette Ban: Who Switches, by Stock



- The response grows with the cigarette stock: smoking-only falls by up to 86 p.p. near the ceiling, mostly to abstinence.
- Vaping-only rises with the cigarette stock, to about 13 p.p. near the ceiling.

Note: Changes 24 months after the ban, by the stocks at the time of the ban, against the baseline at month 24.

Cigarette Ban: E-Cigarette Quantity, by Stock



- E-cigarette quantity rises most among the most heavily smoking-addicted users, to about 12 times its baseline level.
- By the vaping stock it rises most, by about 80%, among households with no vaping stock.

Who Switches Under a Ban

Action after 24 months of the ban (% of the baseline action)

Ban	Baseline action	Share (%)	Abstinence	Smoking-only	Vaping-only
E-cigarettes	Vaping-only	0.72	67.2	32.8	–
	Dual-use	0.13	33.9	66.1	–
Cigarettes	Smoking-only	4.50	90.7	–	9.3
	Dual-use	0.13	62.8	–	37.2

- **Asymmetry.** A third of vapers move to smoking-only, but only 9% of smokers move to vaping-only.
- The gap is smaller than in the first month (13% against 1%). Without cigarettes the cigarette stock decays, so vaping alone becomes attractive to former smokers. Former vapers who smoke build up their smoking stocks.

first month

Removing the Cigarette Excise Tax ($1.84 per Pack)

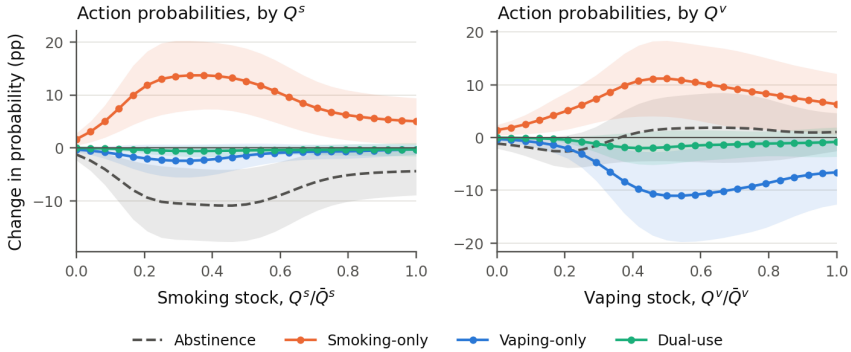
Action shares after 24 months (% of households)

	Baseline	E-cig. ban	Cig. ban	No cig. tax	No e-cig. tax
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Smoking-only	4.50	4.83	–	5.72	4.64
Vaping-only	0.72	–	1.21	0.46	0.84
Dual-use	0.13	–	–	0.09	0.15

- Smoking-only rises from 4.50% to 5.72%. Cigarette quantity +113%, mostly through more packs per smoker (+69%).
- Vaping-only falls from 0.72% to 0.46%, and e-cigarette quantity by 44%.

first month purchase sizes by action

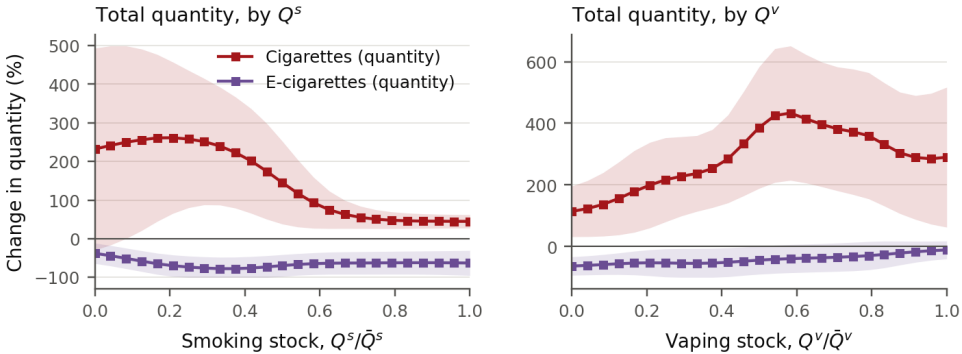
Removing the Cigarette Tax: Choices by Stock



- Smoking-only rises by up to 14 p.p. between a quarter and half of the cigarette-stock ceiling, drawn from abstinence.
- By the vaping stock, vaping-only falls by about 11 p.p. around half the ceiling: addicted vapers switch to smoking-only.

Note: Changes 24 months after the removal, by the stocks at the time of the removal, against the baseline at month 24.

Removing the Cigarette Tax: Quantities by Stock



- Cigarette quantity rises by about 250% below a third of the cigarette-stock ceiling, against about 45% among the most heavily smoking-addicted users.
- E-cigarette quantity falls at every stock.

Removing the E-Cigarette Excise Tax ($0.31 per mL)

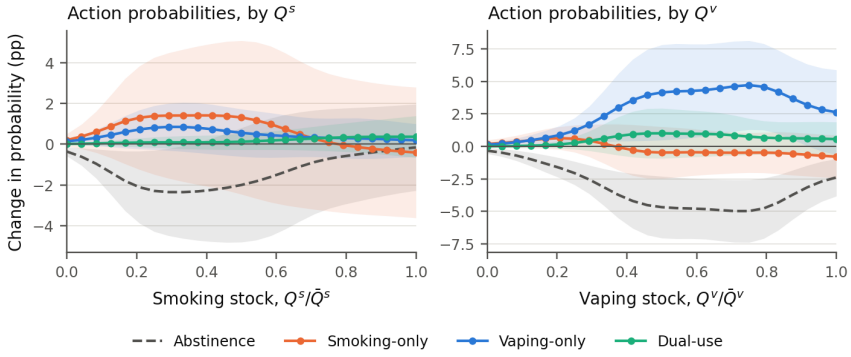
Action shares after 24 months (% of households)

	Baseline	E-cig. ban	Cig. ban	No cig. tax	No e-cig. tax
Abstinence	94.65	95.17	98.79	93.73	94.38
Smoking-only	4.50	4.83	–	5.72	4.64
Vaping-only	0.72	–	1.21	0.46	0.84
Dual-use	0.13	–	–	0.09	0.15

- Vaping-only rises from 0.72% to 0.84%, and e-cigarette quantity by 34%.
- Cigarette quantity also rises (+5.1%): cheaper e-cigarettes do not lower smoking.

first month purchase sizes by action

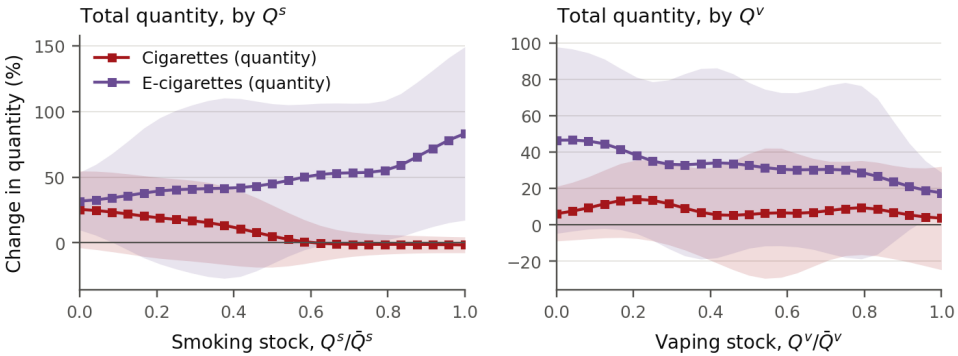
Removing the E-Cigarette Tax: Choices by Stock



- Vaping-only rises by up to about 5 p.p. around three quarters of the vaping-stock ceiling, drawn from abstinence.

Note: Changes 24 months after the removal, by the stocks at the time of the removal, against the baseline at month 24.

Removing the E-Cigarette Tax: Quantities by Stock



- E-cigarette quantity rises with the cigarette stock, from about 30% to about 80% near the ceiling: heavily smoking-addicted users who vape respond strongly to cheaper e-liquid.
- Cigarette quantity rises by up to 25% among lightly smoking-addicted users and fades to zero above about 60% of the ceiling.

What This Means for E-Cigarette Taxes

When e-cigarettes become more expensive, or are banned:

Previous smoker
Moderately smoking-addicted users smoke more.

Previous dual-user
Under a ban, two thirds keep smoking.

Previous vaper
Under a ban, two thirds stop and one third smoke.

- **Asymmetry.** An e-cigarette ban sends a third of vapers to smoking-only. A cigarette ban sends 91% of smokers to abstinence.
- **Effects grow over time.** A 5% e-cigarette tax raises cigarette quantity by 6% at once and by 29% after two years, as the cigarette stocks of the households that switch build up.
- **Addiction profile matters.** Own-price responses fade with the own stock, and substitution comes from moderately addicted users. Policies that account for addiction profiles can target the households a tax actually moves.

Summary

- A dynamic discrete–continuous model of cigarette and e-cigarette demand, with two product-specific addiction stocks, estimated by maximum likelihood using a surrogate network.
- Addiction raises the utility of smoking and vaping up to moderate stocks. For cigarettes it acts mainly on quantity at high stocks: the quantity shifter at the stock ceiling is 11 times that of a new smoker.
- Cigarettes and e-cigarettes are substitutes (cross elasticities 1.3 and 0.6). Own-price responses weaken with addiction.
- After two years of an e-cigarette ban, a third of vapers smoke and cigarette consumption is 1.5% higher. After two years of a cigarette ban, 91% of smokers abstain and e-cigarette consumption is 45% higher.
- A 5% e-cigarette tax raises cigarette quantity by 6% at once and by 29% after two years.

Limitations and Next Steps

Limitations.

- Retention rates are calibrated ($\delta_s = 0.6$, $\delta_v = 0.7$, habit half-lives of about 1.4 and 1.9 months); results by stock depend on this scale.
- The fit at high cigarette stocks is weak, and purchase sizes are slightly under-predicted.

Next steps.

- Welfare measures and the implied optimal cigarette and e-cigarette taxes.
- Counterfactuals with addiction-specific tax levels (optimal tax design by profile).
- A profile likelihood over the retention rates (δ_s, δ_v).

References I

Rahi Abouk, Charles Courtemanche, Dhaval Dave, Bo Feng, Abigail S. Friedman, Johanna Catherine Maclean, Michael F. Pesko, Joseph J. Sabia, and Samuel Safford. Intended and unintended effects of e-cigarette taxes on youth tobacco use. *Journal of Health Economics*, 87:102720, 2023.

Hunt Allcott and Charlie Rafkin. Optimal regulation of e-cigarettes: Theory and evidence. *American Economic Journal: Economic Policy*, 14(4):1–50, 2022. doi: 10.1257/pol.20200805.

Hunt Allcott, Benjamin B. Lockwood, and Dmitry Taubinsky. Regressive sin taxes, with an application to the optimal soda tax. *Quarterly Journal of Economics*, 134(3):1557–1626, 2019. doi: 10.1093/qje/qjz017.

Marlon Azinovic, Luca Gaegauf, and Simon Scheidegger. Deep equilibrium nets. *International Economic Review*, 63(4):1471–1525, 2022.

Mikael Bask and Maria Melkersson. Should one use smokeless tobacco in smoking cessation programs? A rational addiction approach. *European Journal of Health Economics*, 4(4):263–270, 2003. doi: 10.1007/s10198-003-0197-y.

Gary S. Becker and Kevin M. Murphy. A theory of rational addiction. *Journal of Political Economy*, 96(4): 675–700, 1988. doi: 10.1086/261558.

CDC. Smoking and tobacco use: Fast facts. `https://www.cdc.gov/tobacco/data_statistics/fact_sheets/fast_facts/`, 2020. Accessed 2025.

CDC. Health and economic benefits of tobacco use interventions. `https://www.cdc.gov/nccdphp/priorities/tobacco-use.html`, 2024. Published July 12, 2024.

References II

Frank J. Chaloupka. Rational addictive behavior and cigarette smoking. *Journal of Political Economy*, 99(4): 722–742, 1991. doi: 10.1086/261776.

Frank J. Chaloupka and Kenneth E. Warner. The economics of smoking. In Anthony J. Culyer and Joseph P. Newhouse, editors, *Handbook of Health Economics*, volume 1, pages 1539–1627. Elsevier, 2000. doi: 10.1016/S1574-0064(00)80042-6.

Hui Chen, Antoine Didisheim, and Simon Scheidegger. Deep surrogates for finance: With an application to option pricing. *Journal of Financial Economics*, 177:104222, 2026. doi: 10.1016/j.jfineco.2025.104222.

Jialie Chen and Vithala R. Rao. A dynamic model of rational addiction with stockpiling and learning: An empirical examination of e-cigarettes. *Management Science*, 66(12):5886–5905, 2020. doi: 10.1287/mnsc.2019.3490.

Monica E. Cornelius, Teresa W. Wang, Ahmed Jamal, Caitlin G. Loretan, and Linda J. Neff. Tobacco product use among adults — United States, 2019. *MMWR Morbidity and Mortality Weekly Report*, 69 (46):1736–1742, 2020. doi: 10.15585/mmwr.mm6946a4.

Chad D. Cotti, Erik T. Nesson, Michael F. Pesko, Serena Phillips, and Nathan W. Tefft. Standardising the measurement of e-cigarette taxes in the USA, 2010–2020. *Tobacco Control*, 32(e2):e251–e254, 2023. doi: 10.1136/tobaccocontrol-2021-056865.

Chad D. Cotti, Erik T. Nesson, Michael F. Pesko, and Serena Phillips. Standardising the measurement of e-cigarette tax rates in the USA (2nd edition), 2010–2023. *Tobacco Control*, 35(2):173–178, 2026. doi: 10.1136/tc-2024-058618.

References III

John G. Cragg. Some statistical models for limited dependent variables with application to the demand for durable goods. *Econometrica*, 39(5):829–844, 1971.

Jeffrey A. Dubin and Daniel L. McFadden. An econometric analysis of residential electric appliance holdings and consumption. *Econometrica*, 52(2):345–362, 1984.

Tülin Erdem, Susumu Imai, and Michael P. Keane. Brand and quantity choice dynamics under price uncertainty. *Quantitative Marketing and Economics*, 1(1):5–64, 2003.

Emmanuel Farhi and Xavier Gabaix. Optimal taxation with behavioral agents. *American Economic Review*, 110(1):298–336, 2020. doi: 10.1257/aer.20151079.

Aleksandra Friedl, Felix Kübler, Simon Scheidegger, and Takafumi Usui. Deep uncertainty quantification: With an application to integrated assessment models. Working paper, 2023.

Brett R. Gordon and Baohong Sun. A dynamic model of rational addiction: Evaluating cigarette taxes. *Marketing Science*, 34(3):452–470, 2015. doi: 10.1287/mksc.2014.0885.

Jonathan Gruber and Botond Kőszegi. Is addiction "rational"? Theory and evidence. *Quarterly Journal of Economics*, 116(4):1261–1303, 2001. doi: 10.1162/003355301753265570.

Jonathan Gruber and Botond Kőszegi. Tax incidence when individuals are time-inconsistent: The case of cigarette excise taxes. *Journal of Public Economics*, 88(9–10):1959–1987, 2004. doi: 10.1016/j.jpubeco.2003.06.001.

References IV

Rong Hai and James J. Heckman. The causal effects of youth cigarette addiction and education. *Journal of Political Economy*, 134(5):1347–1400, 2026. doi: 10.1086/739828.

Peter Hajek, Anna Phillips-Waller, Dunja Przulj, Francesca Pesola, Katie Myers Smith, Natalie Bisal, Jinshuo Li, Steve Parrott, Peter Sasieni, Lynne Dawkins, Louise Ross, Maciej Goniewicz, Qi Wu, and Hayden J. McRobbie. A randomized trial of e-cigarettes versus nicotine-replacement therapy. *New England Journal of Medicine*, 380(7):629–637, 2019. doi: 10.1056/NEJMoa1808779.

W. Michael Hanemann. Discrete/continuous models of consumer demand. *Econometrica*, 52(3):541–561, 1984.

Igal Hendel and Aviv Nevo. Measuring the implications of sales and consumer inventory behavior. *Econometrica*, 74(6):1637–1673, 2006.

Andrew M. Jones. A double-hurdle model of cigarette consumption. *Journal of Applied Econometrics*, 4(1): 23–39, 1989.

Hanno Kase, Leonardo Melosi, and Matthias Rottner. Estimating nonlinear heterogeneous agents models with neural networks. CEPR Discussion Paper 17391, Centre for Economic Policy Research, 2022.

Lilia Maliar, Serguei Maliar, and Pablo Winant. Deep learning for solving dynamic economic models. *Journal of Monetary Economics*, 122:76–101, 2021.

References V

Ann McNeill, Leonie S. Brose, Ruth Calder, Sarah C. Hitchman, Peter Hajek, and Hayden McRobbie. E-cigarettes: An evidence update. Technical report, Public Health England, London, UK, 2015. "the current best estimate is that using e-cigarettes is around 95% less harmful than smoking".

Jorge J. Moré and Stefan M. Wild. Estimating derivatives of noisy simulations. *ACM Transactions on Mathematical Software*, 38(3):1–21, 2012. doi: 10.1145/2168773.2168777.

National Academies of Sciences, Engineering, and Medicine. *Public Health Consequences of E-Cigarettes*. The National Academies Press, Washington, DC, 2018. doi: 10.17226/24952.

NielsenIQ. Consumer panel data, 2019–2024. Kilts Center for Marketing Data Center, The University of Chicago Booth School of Business, 2024a.

NielsenIQ. Retail scanner data, 2019–2024. Kilts Center for Marketing Data Center, The University of Chicago Booth School of Business, 2024b.

Andriy Norets. Estimation of dynamic discrete choice models using artificial neural network approximations. *Econometric Reviews*, 31(1):84–106, 2012.

Ted O'Donoghue and Matthew Rabin. Optimal sin taxes. *Journal of Public Economics*, 90(10–11): 1825–1849, 2006. doi: 10.1016/j.jpubeco.2006.03.001.

Orzechowski and Walker. The tax burden on tobacco: Historical compilation. Redistributed by the Centers for Disease Control and Prevention (CDC) STATE System, 2024.

References VI

Michael F. Pesko and Casey Warman. Re-exploring the early relationship between teenage cigarette and e-cigarette use using price and tax changes. *Health Economics*, 31(1):137–153, 2022. doi: 10.1002/hec.4439.

Michael F. Pesko, Charles J. Courtemanche, and Johanna Catherine Maclean. The effects of traditional cigarette and e-cigarette tax rates on adult tobacco product use. *Journal of Risk and Uncertainty*, 60(3): 229–258, 2020. doi: 10.1007/s11166-020-09330-9.

Pierpaolo Pierani and Silvia Tiezzi. Addiction and interaction between alcohol and tobacco consumption. *Empirical Economics*, 37(1):1–23, 2009. doi: 10.1007/s00181-008-0220-3.

Henry Saffer, Daniel L. Dench, Michael Grossman, and Dhaval M. Dave. E-cigarettes and adult smoking: Evidence from Minnesota. *Journal of Risk and Uncertainty*, 60(3):207–228, 2020. doi: 10.1007/s11166-020-09326-5.

Simon Scheidegger. Deep learning for solving and estimating dynamic models in economics and finance. arXiv:2605.14493, 2026.

World Health Organization. *WHO Report on the Global Tobacco Epidemic, 2015: Raising Taxes on Tobacco*. World Health Organization, Geneva, 2015.

Appendix: Identification and Calibration

- Curvature (κ_0, κ_1) and money α: from how conditional quantities respond to prices.
- Stock terms in f and shifters ψ: participation and quantity jointly, and how both move as stocks accumulate within a household.
- Cross-stock terms: dual-use and switches between products.
- Price sensitivity: tax-driven price variation across states and price innovations orthogonal to the projected demand shocks, after the control function. Exclusion: taxes affect utility only through prices.
- Control function: $\boldsymbol{\lambda}_v$ from how vaping moves with the current residual $\widehat{\boldsymbol{v}}_{ct}$; b_v from whether it also shifts conditional quantities.
- Types: the first months of each household are the most informative (the seed gap decays at δ_s, indistinguishable after about six months); longer sequences separate the seed from the permanent intercepts c_τ.

Calibrated: $\beta = 0.95$; $\delta_s = 0.6$, $\delta_v = 0.7$ how; $\rho_v = 0.83$ (poorly identified when the loadings are small).

back

Appendix: Sample Construction

	Households	Household-months
Single-person households, 2019–2024	27,780	1,100,736
head of household under 50	6,779	189,845
price observed, complete demographics, states with $>$ 30 households	**6,538**	**183,147**

1,214 counties in 38 states; the median household is observed for 23 months. Single-person households so that purchases can be attributed to one person; under 50 because vaping is rare among older adults.

back

Appendix: Latent Types

	Households	τ_0 no prior use	τ_s prior smoker	τ_v prior vaper	τ_{sv} prior dual-user
A. Prior type probabilities (%)					
All households	6,538	77.5	6.7	5.9	9.9
Under $25K	1,608	65.5	**11.3**	8.1	**15.1**
$25–45K	1,664	73.8	5.8	**9.3**	11.0
$45–70K	1,642	85.2	5.2	2.4	7.2
$70–100K	985	86.5	3.8	3.3	6.5
$100K+	639	89.5	3.0	3.3	4.3
B. Action intercepts $c_{\tau,a}$					
Smoking-only		−11.53	**−4.14**	−7.75	**−5.88**
Vaping-only		−19.17	**−9.89**	**−10.33**	−13.31
Dual-use		−23.02	−10.96	−15.44	−15.26

- Low-income households more often enter with high stocks.
- Prior smokers have the highest utility of smoking-only, and also of vaping-only.

back

Appendix: Types vs. Actions

Symbol	Description	Cigarette component	E-cigarette component
Panel A. Latent type τ (initial stock at panel entry)			
τ_0	no prior use	low	low
τ_s	prior smoker	high	low
τ_v	prior vaper	low	high
τ_{sv}	prior dual-user	high	high
Panel B. Current action a (any month)			
$a = 0$	abstinence	inactive	inactive
$a = s$	smoking-only	active	inactive
$a = v$	vaping-only	inactive	active
$a = sv$	dual-use	active	active

High seed: $Q^g_{H,0} = \min\{\bar{Q}_g,\ P_{75}(\bar{\chi}^g_i \mid \bar{\chi}^g_i > 0)/(1-\delta_g)\}$, where $\bar{\chi}^g_i$ is the household's average monthly stock increment over its observed months. Low seed: 0.

Type logit covariates $\boldsymbol{z}_i$: intercept, demographic indicators (gender, marital status, age group, education, occupation, race, income band) and income $\times$ education interactions. A small ridge penalty keeps the slopes finite.

back

Appendix: Control Function — Assumptions

Let ξ_{ct} shift participation and $\xi^{int}_{ct} = b\,\xi_{ct}$ shift $\log \Phi$ (the quantity margin). $\boldsymbol{\tau}_{st}$ are state excise taxes.

1. **Persistence.** $\xi_{ct} = \rho\,\xi_{c,t-1} + \eta_{ct}$ with $\mathbb{E}[\eta_{ct} \mid \boldsymbol{\tau}_{st}, \mathcal{I}_{c,t-1}] = 0$.
2. **Pricing.** Firms observe ξ_{ct} when setting prices.
3. **Distribution.** Firm-observed demand and supply innovations are jointly normal.

Taxes may respond to persistent prevalence, but not to this month's innovation.

first stage back

Appendix: Control Function — First Stage and Construction

First stage (county-month prices on taxes, with county and time effects):

$$\boldsymbol{p}_{ct} = B_0 \boldsymbol{\tau}_{st} + B_1 \boldsymbol{\tau}_{s,t-1} + \textstyle\sum_{\ell=1}^{L} B_{2,\ell} \boldsymbol{p}_{c,t-\ell} + \boldsymbol{\mu}_c + \boldsymbol{\zeta}_t + \boldsymbol{v}_{ct}$$

Projecting ξ on $\boldsymbol{v}$ and iterating the AR(1) backwards splits the shock into a part correlated with the price history and an orthogonal remainder:

$$\xi_{ct} = \underbrace{C_{ct}}_{\text{control function}} + \underbrace{\mathcal{E}_{ct}}_{\perp \text{ prices}}, \qquad C_{ct} = \rho\, C_{c,t-1} + \boldsymbol{\lambda}' \boldsymbol{v}_{ct}.$$

assumptions back

Appendix: Control Function Estimates

$$u_{it}(x) = \bar{u}_{it}(x) + \widehat{C}^{v}_{ct} + \varepsilon_{it} \qquad \text{(vaping)}$$

$$\log \Phi^{v}_{it} = \psi_{0v} + \psi_{v} Q^{v}_{it} + b_{v} \widehat{C}^{v}_{ct}$$

$$\widehat{C}^{v}_{ct} = \rho_{v} \widehat{C}^{v}_{c,t-1} + \lambda_{vs} \hat{v}^{s}_{ct} + \lambda_{vv} \hat{v}^{v}_{ct}, \qquad \widehat{C}^{s}_{ct} \equiv 0, \quad \rho_{v} = 0.83 \text{ (calibrated)}$$

Control function.

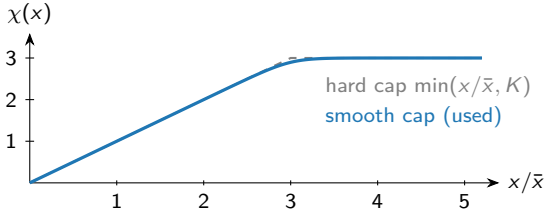
- The loadings of the e-cigarette control function are small and not jointly significant: $\lambda_{vs} = 0.004$ (0.040), $\lambda_{vv} = 0.050$ (0.045), $p = 0.46$. Its coefficient in the e-cigarette shifter is insignificant too: $b_{v} = 0.29$ (0.26).
- Once the stocks and the year-month and county effects are in, local demand shocks correlated with e-cigarette price innovations play little role. The cigarette loadings were very close to zero and dropped.

back: control function estimates

Appendix: How the Stock Increment $\chi(x)$ Is Computed

In estimation the increment is a smoothed version of $\min(x/\bar{x},\, K)$:

$$\chi(x) = K - \varepsilon \log\!\left(1 + \exp\left(\frac{K - x/\bar{x}}{\varepsilon}\right)\right) \quad (x > 0), \qquad \chi(0) = 0, \qquad K = 3, \;\; \varepsilon = 0.05K.$$



- $\bar{x}$ is the median purchase (about 4 packs, 16 mL or 8 cartridges): a median purchase adds one unit, no purchase adds nothing.
- The cap $K = 3$ stops bulk buys from creating implausibly large stocks.
- Smoothing: $\chi \approx x/\bar{x}$ below the cap and $\chi \to K$ above it. The slope $\chi'(x) = \sigma\big((K - x/\bar{x})/\varepsilon\big)/\bar{x}$ falls smoothly from $1/\bar{x}$ to 0, so the first-order condition has a derivative everywhere; the hard min has a kink at the cap.

back

Appendix: Calibrating the Retention Rates

Idea. If past purchases matter only through a geometric stock, a purchase k months ago matters in proportion to δ_g^{k-1}:

$$d_{it} = \sum_{k=1}^{6} b_{-k}\, d_{i,t-k} + b_p\, p_t + \text{controls} + e_{it}, \qquad b_{-k} = \lambda\, \delta_g^{k-1},$$

so the decay of the lag coefficients estimates δ_g.

	lag 1	lag 2	lag 3	lag 4	lag 5	lag 6
Smoke	0.28	0.19	0.12	0.10	0.10	0.10
Vape	0.25	0.19	0.13	0.11	0.08	0.09

- Lags 1–4 decay geometrically; deep lags flatten at a floor (persistent heterogeneity, not habit).
- Across specifications $\delta_s \in [0.46, 0.69]$ and $\delta_v \in [0.62, 0.76]$; selected $\delta_s = 0.6$, $\delta_v = 0.7$.

Note: Linear probability of a purchase on six monthly lags, own price, demographics and time effects; each household's first ten months dropped; pooled and household-demeaned. Follows Hai and Heckman (2026).

back

Appendix: Prices and Beliefs

- **Prices.** Real, tax-inclusive county-month price indices per pack and per mL from the retail scanner data, composition-adjusted across products and stores. Households in the same county-month face the same prices.

- **Beliefs.** Each price follows an AR(1) estimated in a first step and held fixed:

$$p^g_{c,t+1} = \mu_g + a_g\, p^g_{ct} + r^g_{c,t+1}.$$

  The price innovations $\boldsymbol{r}$ are drawn jointly with the first-stage innovations $\boldsymbol{v}$, so a demand innovation moves next month's price and control function together.

- **First stage.** Prices on current and lagged excise taxes, own lags, county and month effects; the residual $\widehat{\boldsymbol{v}}_{ct}$ builds $C^g_{ct} = \rho_g C^g_{c,t-1} + \boldsymbol{\lambda}'_g \widehat{\boldsymbol{v}}_{ct}$.

back

Appendix: Standard Errors

- The surrogate gives **exact gradients** of each household's log-likelihood contribution: the scores s_i.
- Sandwich with household clusters and weights w_i:

$$\widehat{\mathrm{Var}}(\hat{\theta}) = B^{-1} M B^{-1}, \qquad B = \sum_i w_i \, s_i s_i', \qquad M = \sum_i (w_i s_i)(w_i s_i)'.$$

- The outer product replaces the Hessian: the differenced Hessian of the network likelihood is not negative definite along the network-input directions.
- Inference is conditional on the first stage, the price process, δ_g and ρ_v.
- **Training box.** The nets are trained once on a fixed box of parameter values; the estimate lies inside it.

back

Appendix: The Value and Policy Nets

The value net $\bar{V}_\varphi(\boldsymbol{S}^+, h; \theta)$ approximates the expected continuation value after the quantity choice. The slope in the FOC is its own derivative (automatic differentiation through next month's stocks):

$$\Lambda(\boldsymbol{S}^+, \theta) = \frac{\partial \bar{V}_\varphi(\boldsymbol{S}^+, h; \theta)}{\partial Q^s_{t+1}},$$

so the policy net is optimal for exactly the value the likelihood uses. The policy net $x_{\mathrm{NN}}(\boldsymbol{S}, \theta)$ is trained to satisfy the first-order condition:

$$R(\boldsymbol{S}, \theta) = \frac{\partial \bar{u}(\boldsymbol{S}, x; \theta)}{\partial x} + \beta\, \mathbb{E}\big[\Lambda(\boldsymbol{S}', \theta) \mid \boldsymbol{S}\big]\, \chi'(x), \qquad x \geq 0, \quad R \leq 0, \quad xR = 0.$$

training details | Fischer–Burmeister loss | back

Appendix: Surrogate Training — What the Nets Learn

- **Networks.** A value net $\bar{V}$ and a policy net $\boldsymbol{x}_{\mathrm{NN}}$. Inputs: the state, the parameters θ, and the index h of type intercepts, demographics and fixed effects.
- **Training points.** Observed, simulated and uniformly drawn states; θ uniform on a box around the $\beta = 0$ estimate; $K = 4$ draws of next month's prices and control function per state.
- **Targets**, from a slowly updated copy $\bar{\varphi}$ of the nets:

$$t_V = \frac{1}{K} \sum_{k=1}^{K} \log \sum_{a} \exp \left\{ \max_{x_a \geq 0} \left[\bar{u}_a(\boldsymbol{S}'_k, x_a; \theta) + \beta \, \bar{V}_{\bar{\varphi}}(\boldsymbol{S}^{+\prime}_{k,a}, h; \theta) \right] \right\}$$

The simulation noise in the K-draw averages washes out across the training sample (Maliar et al., 2021).

losses back

Appendix: Surrogate Training — Losses

- **Value**: squared error scaled by the variance of the target, plus a slope (Sobolev) term so that the derivative used in the FOC is fitted as well,

$$\mathcal{L}_V = \frac{\operatorname{mean}\left[(\bar{V}_\varphi - t_V)^2\right]}{\operatorname{var}(t_V)}.$$

- **Policy**: $\mathcal{L}_X = \operatorname{mean}\left[\phi_{\mathrm{FB}}(a, b)^2\right]$, the first-order condition written as one equation.
- **Total**: $\mathcal{L} = \mathcal{L}_V / n_V + \mathcal{L}_X / n_X$, each term scaled by its level early in training so the two get equal weight.

Fischer–Burmeister loss back

Appendix: Fischer–Burmeister Loss for the Quantity Policy

The optimal quantity solves a **complementarity problem**, not just an equation:

$$x \geq 0, \qquad R \leq 0, \qquad x\,R = 0.$$

Either the household buys and the first-order condition holds ($R = 0$), or it buys nothing because the marginal unit is not worth its price ($x = 0$, $R \leq 0$).

The Fischer–Burmeister function turns these three conditions into one equation:

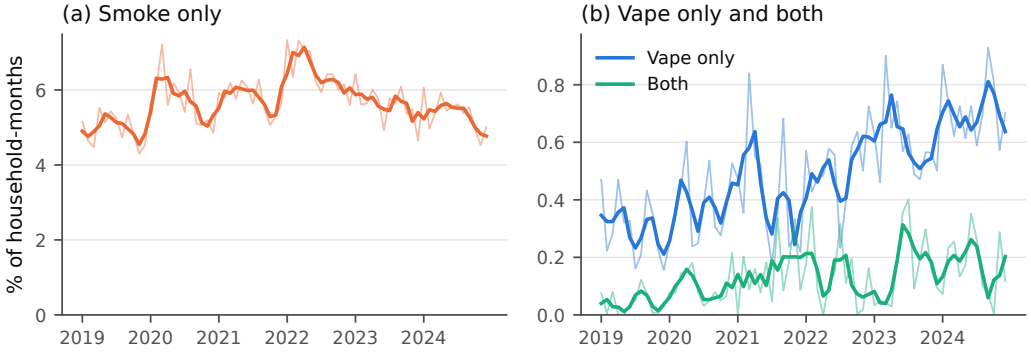
$$\phi_{\mathrm{FB}}(a, b) = a + b - \sqrt{a^2 + b^2} = 0 \quad \Longleftrightarrow \quad a \geq 0, \;\; b \geq 0, \;\; ab = 0.$$

- $a = (x - x_{\min})/\bar{x}$: distance from the lower bound, in units of the median purchase.
- $b = -R/s$, with $s = |\Phi\,\phi'| + |\alpha p| + |\beta\,\text{continuation}|$, so residuals are comparable across products and states.
- The loss $\phi_{\mathrm{FB}}(a, b)^2$ is zero exactly at the optimum, smooth, and needs no switching between interior and corner cases. At the upper cap on quantity with $R > 0$ the residual is set to zero.

The same device handles occasionally binding constraints in deep-learning solution methods (Maliar et al., 2021; Azinovic et al., 2022).

targets and losses back

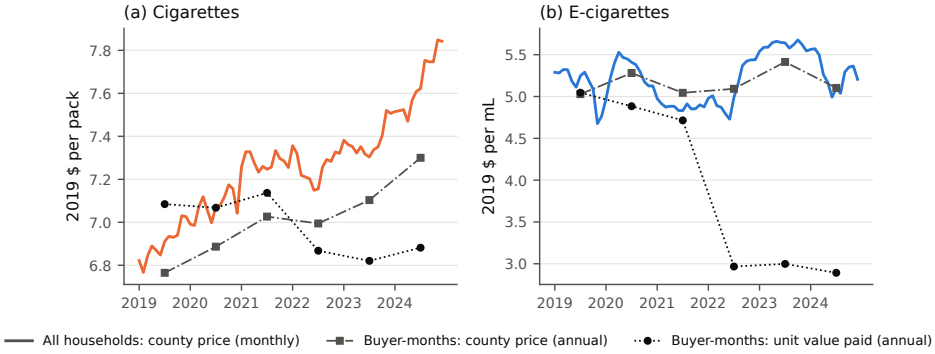
Appendix: Descriptives — Action Shares



- Smoking-only stays between about 4.5% and 7% of household-months, peaking in early 2022.
- Vaping-only roughly doubles over the sample (about 0.3% to 0.7%); dual-use stays below 0.3%.

back

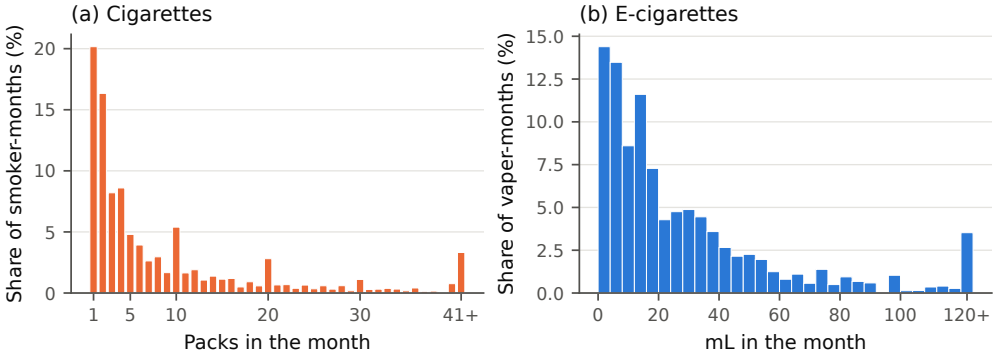
Appendix: Descriptives — Prices



- Cigarettes: the county price rises steadily in real terms; the unit value buyers pay stays near $7 per pack.
- E-cigarettes: the county price moves between $4.7 and $5.7 per mL, but the unit value buyers pay falls from about $5 to $3 in 2022, a shift to cheaper products.

back

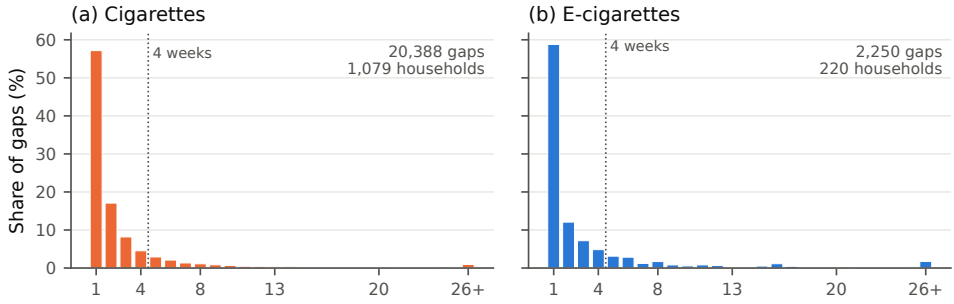
Appendix: Descriptives — Purchase Sizes



- Right-skewed: most smoker-months buy one or two packs, with spikes at cartons (10, 20 packs); most vaper-months buy under 20 mL.
- Quantities among buyers rise over the sample: compositional for cigarettes (which households buy), within-household and compositional for e-cigarettes.

back

Appendix: Descriptives — Time Between Purchases

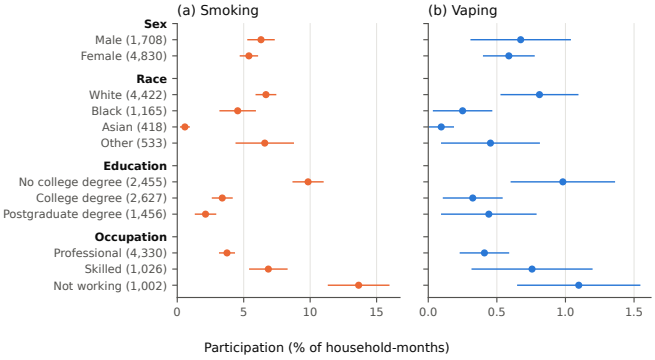


Weeks between consecutive purchase weeks of the same household

- Purchases are frequent: close to 60% of gaps between purchase weeks are a single week, and more than 80% are within four weeks, for both products.

back

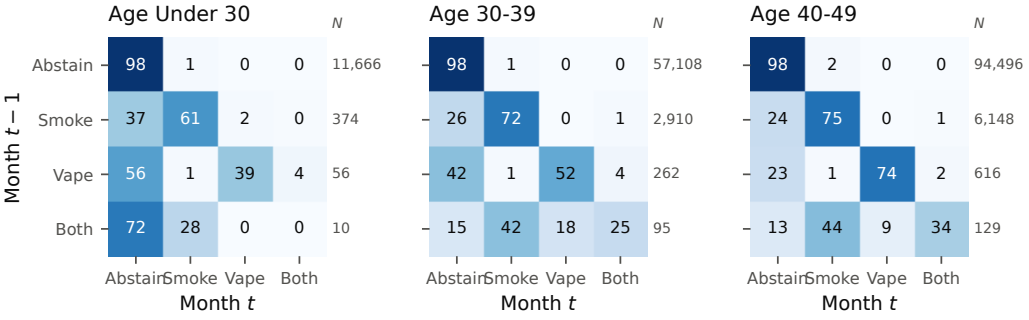
Appendix: Participation by Demographics



- Smoking falls with education and income (11.2% below $25k to 2.4% above $100k) and is highest among those not working.
- Vaping shows little income gradient; it is highest among those without a degree or not working.

back

Appendix: Transitions by Age



- Persistence rises with age: smokers smoke again next month 61% of the time under 30 and 75% at 40–49; vapers 39% and 74%.
- Young vapers mostly stop: 56% of vapers under 30 abstain the next month.

back

Appendix: Under-Recording

- Panel purchase rates (5.8% smoke, 0.63% vape per household-month) are well below NHIS current use (2019: 14.0% and 4.5%; Cornelius et al., 2020).
- Why: not every purchase is recorded; gas stations and corner stores, where much tobacco is bought, are under-covered; and purchases are not consumption.
- A recording model can be written down, but the thin vaping sample cannot identify it, so the estimates describe **recorded** purchases.

recording model back

Appendix: True and Recorded Actions

- True action $a_{it} \in \{0, s, v, sv\}$; recorded action o_{it}. Recording is **one-sided**: a recorded purchase is a true purchase, but a true purchase of product g is recorded only with probability r_k^g (demographic cell k).

Recording probability $M(o \mid a; r)$

$a \backslash o$	0	s	v	sv
0	1	0	0	0
s	$1 - r^s$	r^s	0	0
v	$1 - r^v$	0	r^v	0
sv	$(1 - r^s)(1 - r^v)$	0	0	$1 - (1 - r^s)(1 - r^v)$

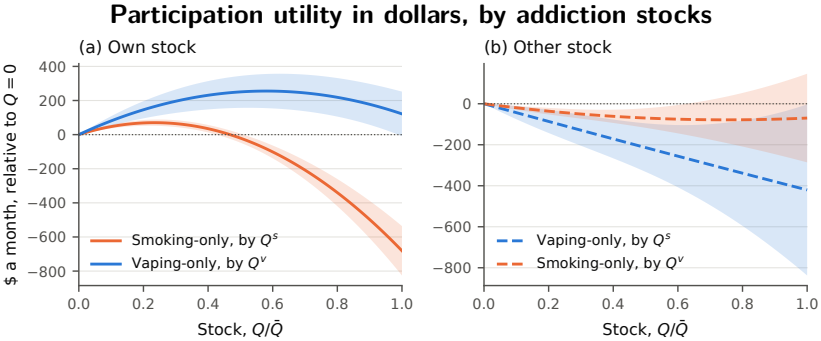
- **Likelihood**: the choice probability of a recorded month becomes

$$P_\tau\big(o_{it} \mid \boldsymbol{S}_{it}^\tau, \boldsymbol{z}_i\big) \;\longrightarrow\; \sum_{a \in \mathcal{A}} M(o_{it} \mid a; r)\, P_\tau\big(a \mid \boldsymbol{S}_{it}^\tau, \boldsymbol{z}_i\big).$$

- Only recorded zeros mix: no recorded purchase is a true abstinence or an unrecorded purchase, weighted by the state, so a high-stock household that records nothing has most likely bought.

back

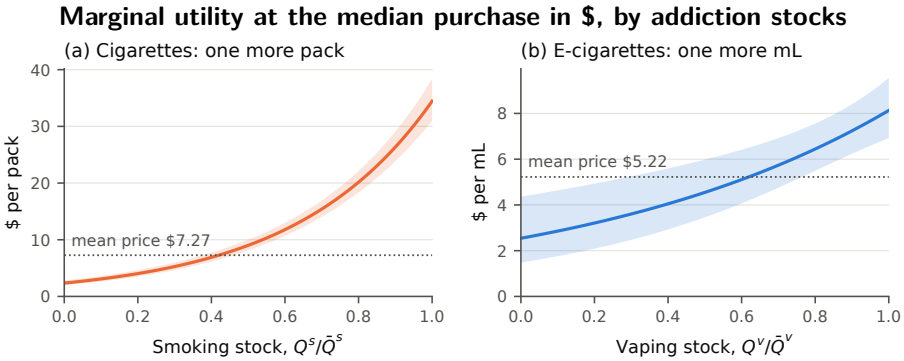
Appendix: Participation Utility by the Addiction Stocks



- **(a) Own stock.** Inverted U: smoking peaks at about \$70 a month at 23% of the ceiling and falls to −\$680 at the ceiling. Vaping stays positive.
- **(b) Other stock.** The cigarette stock lowers the utility of vaping-only throughout. The vaping stock lowers the utility of smoking-only by a small amount.

back

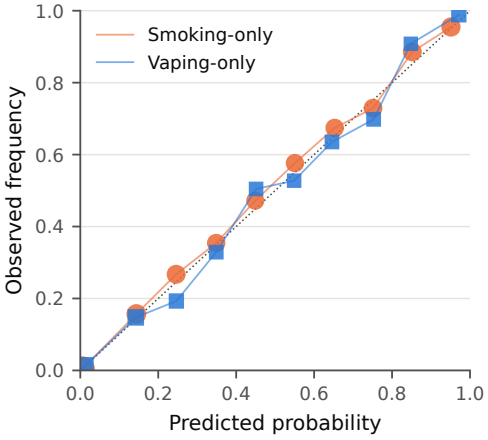
Appendix: Marginal Value of Quantity by Stock



- An additional pack is worth \$2.36 with no cigarette stock, \$9.01 at half the ceiling and \$34.45 at the ceiling (mean price \$7.27).
- An additional mL is worth \$2.54 with no vaping stock and \$8.13 at the ceiling (mean price \$5.22): the shifter falls, but the curvature falls faster.

back

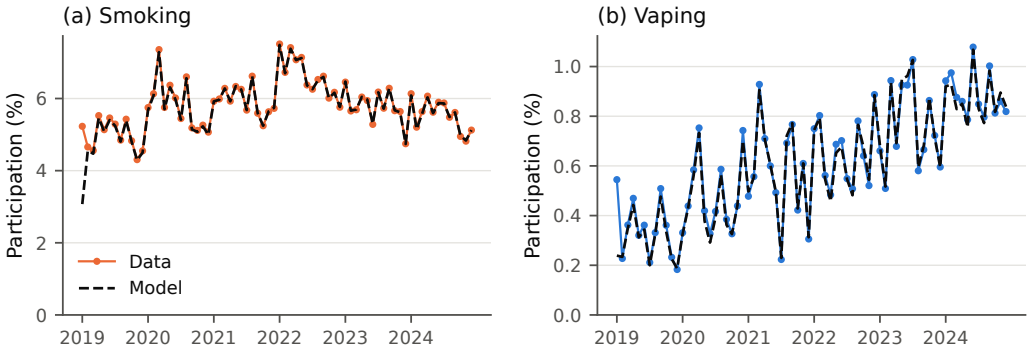
Appendix: Calibration



Note: Observed frequency against predicted probability by bins of the predicted probability (marker size = household-months).

back

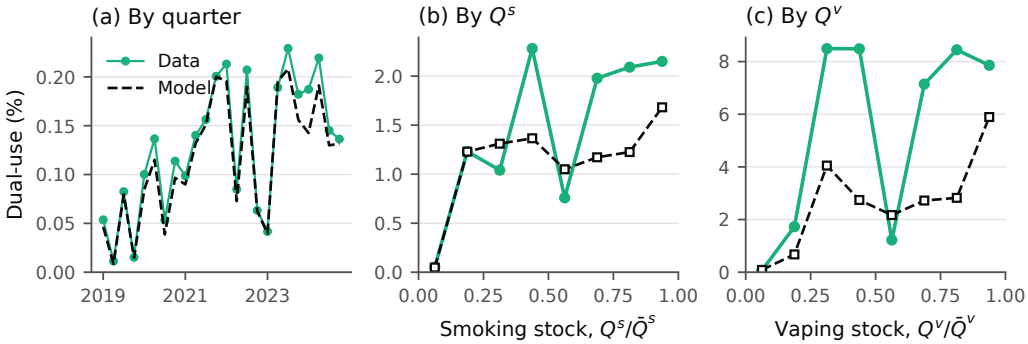
Appendix: Fit by Period



Note: Observed and predicted participation by calendar month. Predicted smoking follows the time pattern, including the 2022 step, within 0.2 p.p. of the data in every year. Predicted vaping rises from 0.31% in 2019 to 0.87% in 2024, against 0.34% and 0.89% observed.

back

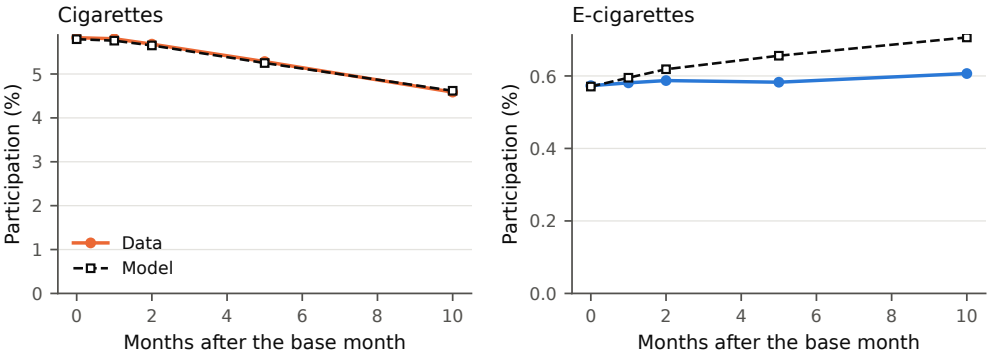
Appendix: Dual-Use



Note: Dual-use is matched overall (0.12% against 0.13%) and in every year (0.04–0.15% against 0.04–0.17%). By the cigarette stock the predictions stay within one p.p. of the data. Among households with high vaping stocks it is under-predicted, on few household-months.

back

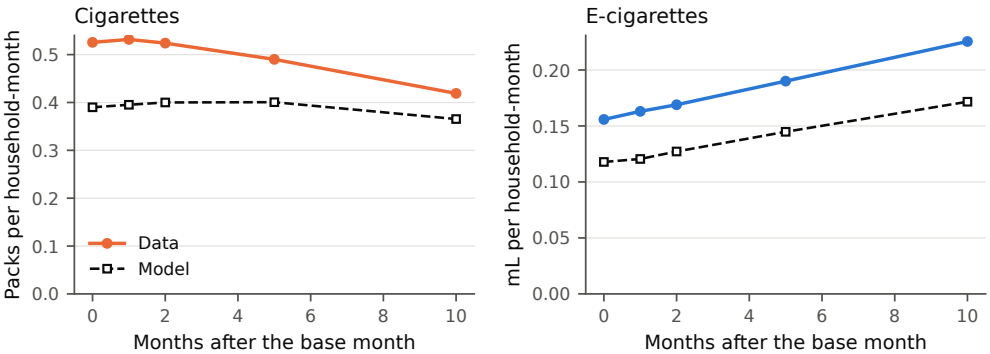
Appendix: Dynamic Fit — Participation



- From each of 62 base months, households are simulated ten months forward along realised prices and compared with the same households in the data.
- The simulation tracks smoking (5.8% to 4.6% over ten months, as in the data). Simulated vaping drifts up from 0.57% to 0.71%, while the data rise to 0.61%.

back

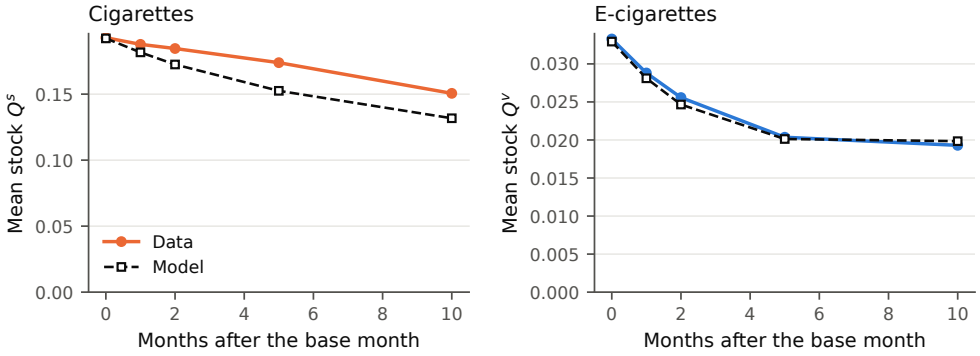
Appendix: Dynamic Fit — Purchases



- Simulated purchases start too low and stay too low: 0.39 against 0.53 packs and 0.12 against 0.16 mL per household-month in the base month.

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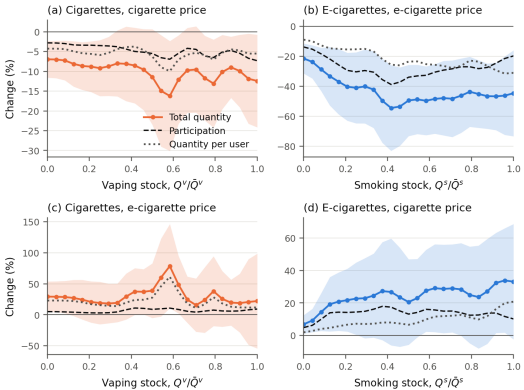
Appendix: Dynamic Fit — Stocks



- The simulated cigarette stock falls behind the data, by 13% after ten months: simulated purchases are too small.
- The simulated e-cigarette stock stays with the data (0.020 against 0.019 after ten months).

back

Appendix: 24-Month Responses by the Other Product's Stock



- A 5% e-cigarette tax lowers e-cigarette quantity most among heavily smoking-addicted users (-45% near the smoking-stock ceiling).
- A 5% cigarette tax raises e-cigarette quantity by 7% with no smoking stock and by 33% near the ceiling: smokers are the ones who move to e-cigarettes.

back

Appendix: Policy Experiments — Simulated Population

- 676,000 simulated households on a grid of cells over the two stocks, 4,000 per cell. Each cell holds households whose demographics and types are distributed as in the data at that addiction level.
- Each household draws its prices and e-cigarette control function from a random observed month of its county. Cells are weighted by the joint distribution of the stocks in the data.
- Pass-through to the consumer price is one; households expect the tax to persist. The removals lie inside the trained tax range.
- Month 0 is the immediate response; then households draw actions and purchases from the model for 24 months, and their stocks evolve.
- The baseline drifts: simulated smoking falls from 5.7% to 5.0% after a year, vaping rises from 0.61% to 0.76%. Effects are measured against this moving baseline.
- Bands: 50 parametric bootstrap draws of the utility parameters.

back

Appendix: Purchase Size by Action After 24 Months

			After 24 months			
Product	Action	Baseline	E-cig. ban	Cig. ban	No cig. tax	No e-cig. tax
Cigarettes (packs)	Smoking-only	8.11	7.81	–	13.67	8.24
	Dual-use	5.31	–	–	9.86	5.96
E-cigarettes (mL)	Vaping-only	30.8	–	30.1	28.1	35.4
	Dual-use	22.5	–	–	13.7	27.8

- Under the e-cigarette ban, packs per smoking-only month fall from 8.11 to 7.81: former vapers who smoke buy fewer packs than existing smokers.
- Removing the cigarette excise raises packs per smoking-only month by 69%.

Note: Mean purchase per month among the households choosing the action at month 24, simulated population. A dash marks an action removed from the choice set.

back: e-cig ban cig ban no e-cig tax

Appendix: Bans and Tax Removals in the First Month

Action shares at month 0 (% of households)

	Baseline	E-cig. ban	Cig. ban	No cig. tax	No e-cig. tax
Abstinence	93.79	94.26	99.44	93.50	93.77
Smoking-only	5.60	5.74	–	5.94	5.58
Vaping-only	0.48	–	0.56	0.45	0.52
Dual-use	0.12	–	–	0.11	0.14

- E-cigarette ban: cigarette quantity does not change (-0.1%): the households that move into smoking-only buy fewer packs.
- Cigarette ban: vaping participation falls by 8% and e-cigarette quantity by 3.7%: the dual-users who stop vaping outnumber the smokers who switch.
- Removing the cigarette excise raises cigarette quantity by 33% at once; removing the e-cigarette excise raises e-cigarette quantity by 17%.

back: e-cig ban | cig ban | no cig tax | no e-cig tax

Appendix: Who Switches in the First Month of a Ban

Action after the ban, month 0 (% of the baseline action)

Ban	Baseline action	Share (%)	Abstinence	Smoking-only	Vaping-only
E-cigarettes	Vaping-only	0.48	87.1	12.9	–
	Dual-use	0.12	39.2	60.8	–
Cigarettes	Smoking-only	5.60	99.0	–	1.0
	Dual-use	0.12	85.4	–	14.6

- At once, 13% of vapers move to smoking-only and 1% of smokers to vaping-only.
- At high cigarette stocks almost every vaping household is a dual-user, and these households stop vaping rather than vape alone.

back

Appendix: Open Issues

- **Quantity fit.** Purchase sizes are under-predicted: 6.9 against 9.3 packs and 23.5 against 29.2 mL per buyer-month.
- **Heavy smokers.** The fit at high cigarette stocks is weak.
- **Simulated baseline drifts.** Starting from the data's stock distribution, simulated smoking falls from 5.7% to 5.0% after a year and vaping rises from 0.61% to 0.76%. Policy effects are measured against this moving baseline.
- **Control-function persistence** ρ_v is calibrated, with small loadings it is weakly identified.
- **Retention rates** calibrated from auxiliary regressions, a profile likelihood over (δ_s, δ_v) is the planned replacement.